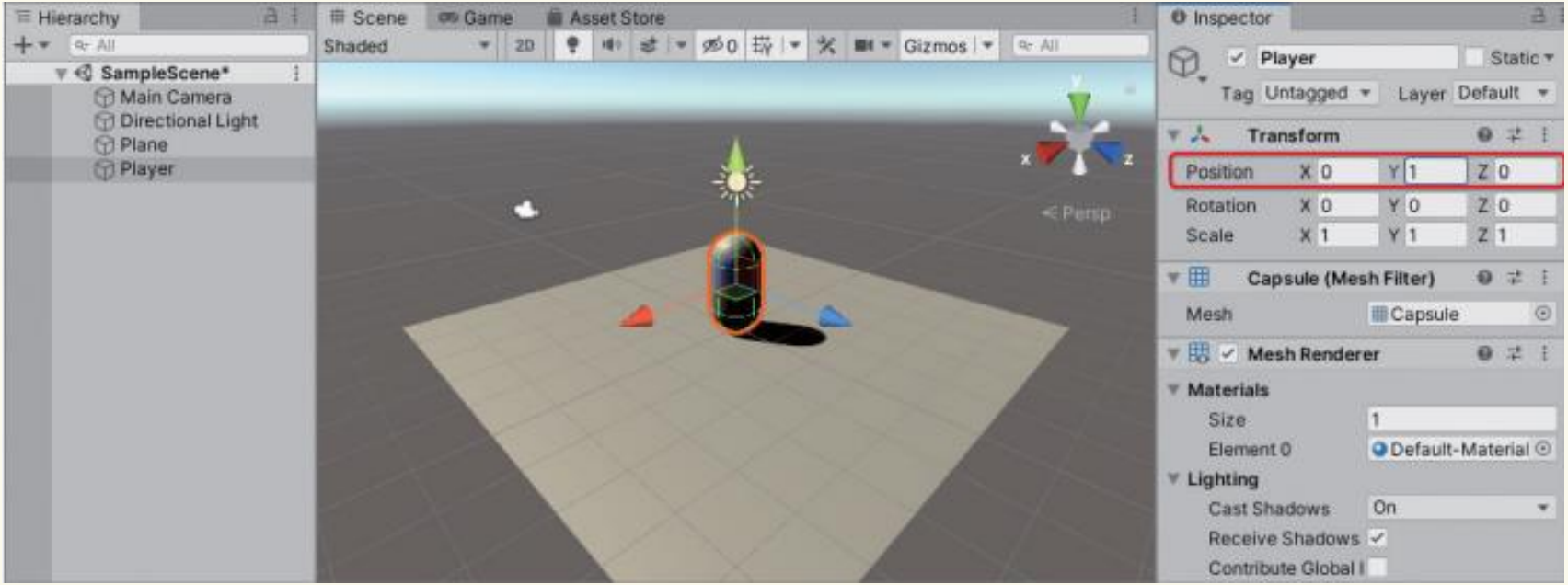
유니티 기초

CODING SOFTWARE CREATIVE EDUCATION CENTER

프리팹과 UI

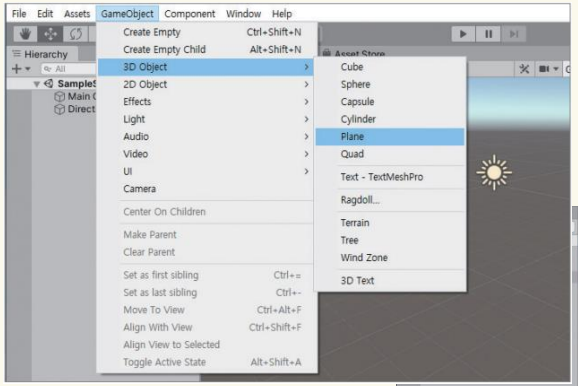
프리팹을 이용한 게임 제작

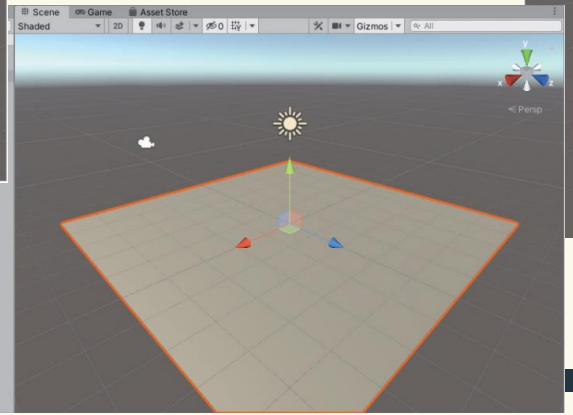
- 프로젝트 생성 및 이동 스크립트 작성: TPS01

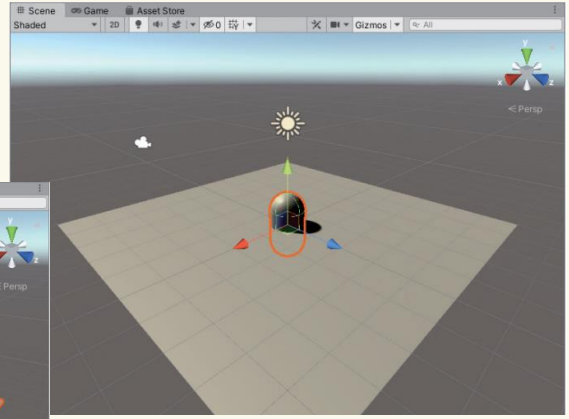


프리팹을 이용한 게임 제작

- 리소스 배치
  - 유니티가 제공하는 기본 3D 오브젝트에서 플레인과 캡슐을 선택
  - 메뉴에서 [GameObject]-[3D Object]-[Plane]을 선택하면 [Scene] 창에 플레인이 나타나고 [Hierarchy] 창에는 플레인 오브젝트가 생성
  - [GameObject]-[3D Object]-[Capsule]을 선택하면 플레인에 캡슐 등장

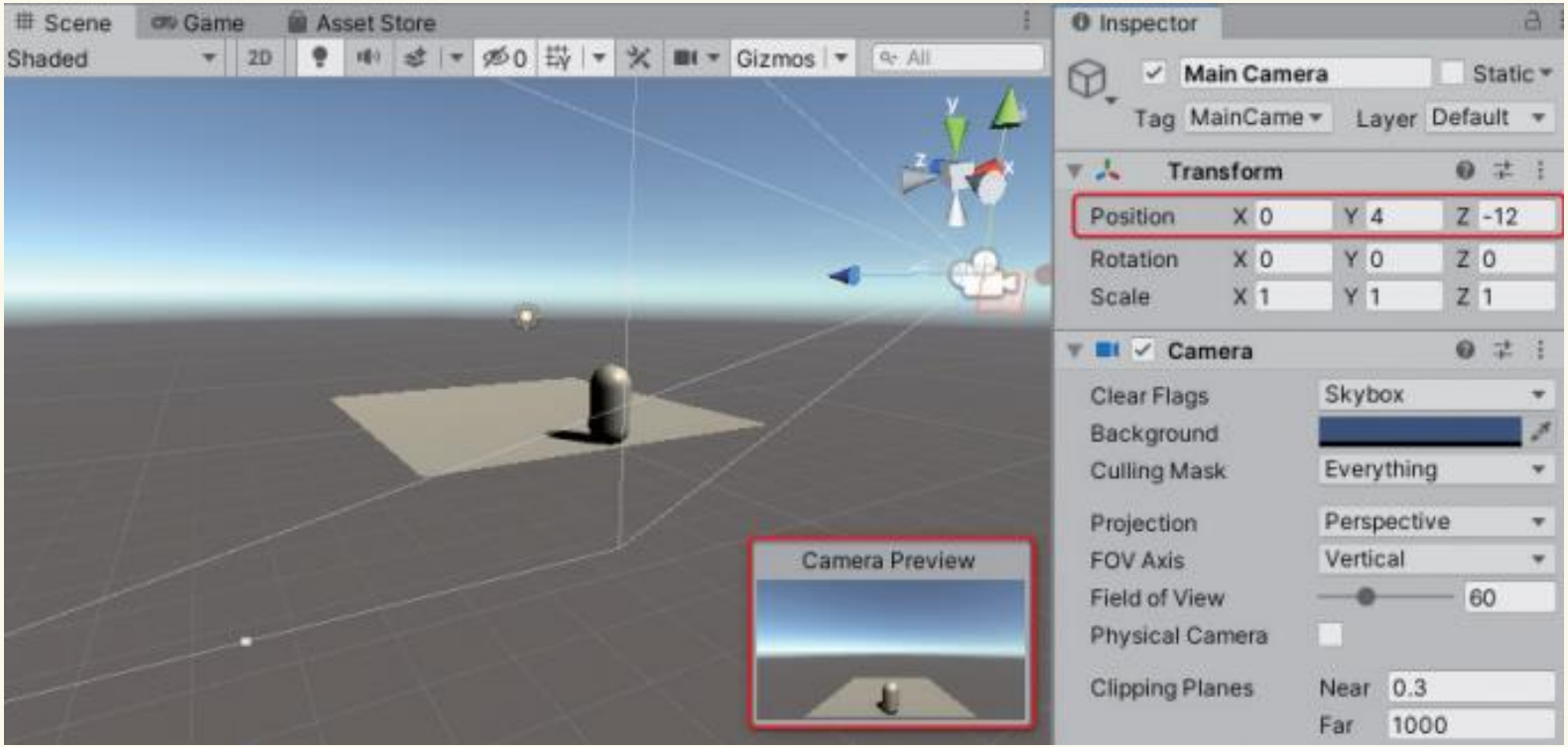






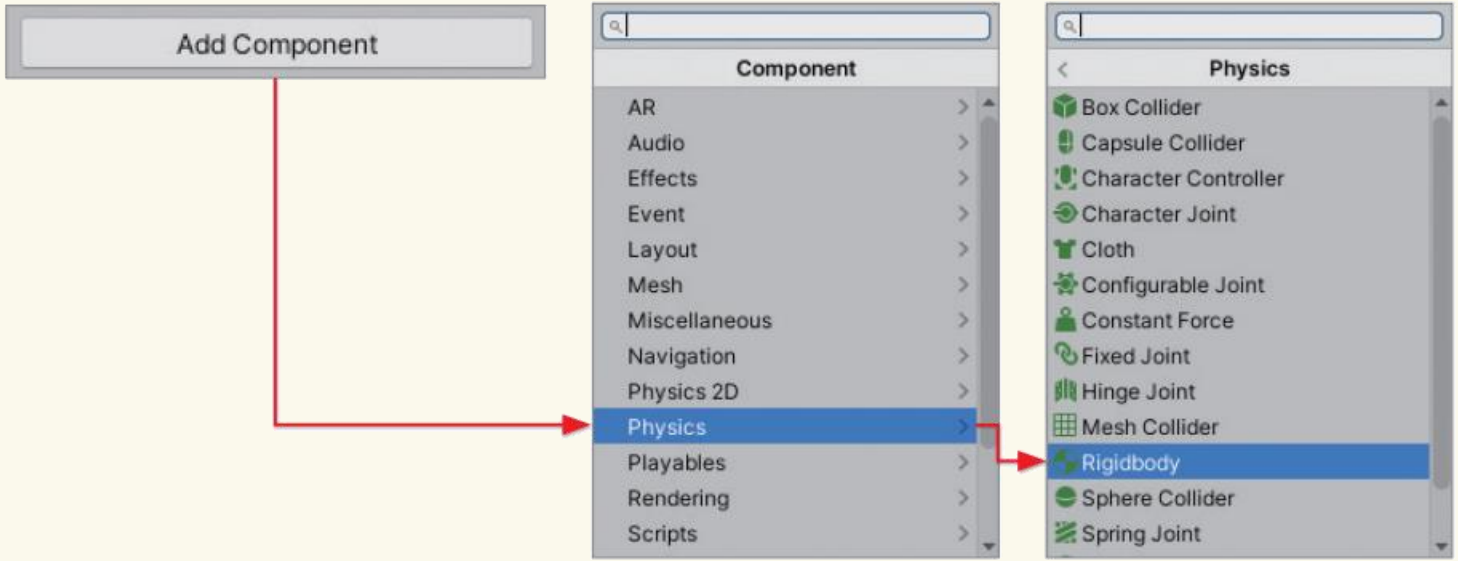
프리팹을 이용한 게임 제작

- 카메라 위치 조정



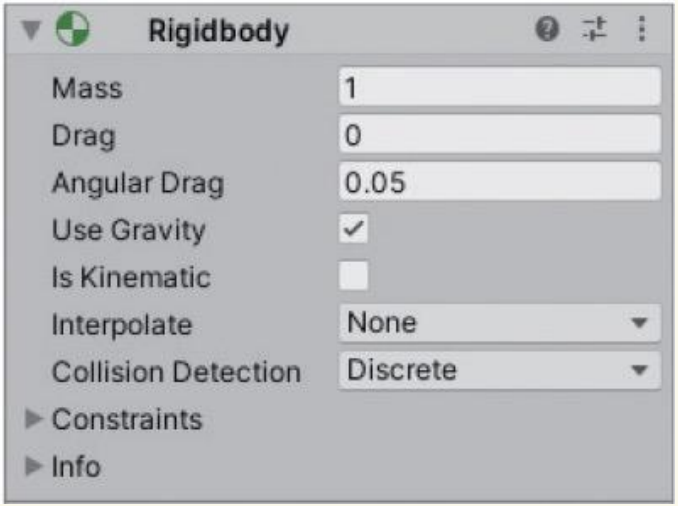
02. 물리 엔진 적용

- Rigidbody 컴포넌트 적용하기
  - [Hierarchy] 창에서 Player를 선택한 후 [Inspector] 창에서 <Add Component>를 클릭하고 [Physics]-[Rigidbody]를 선택

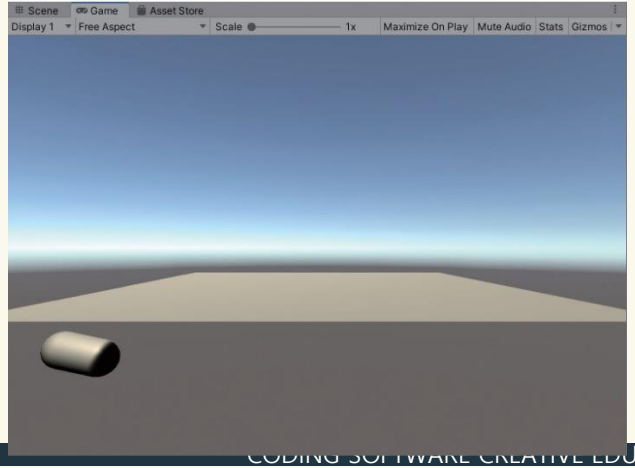


02. 물리 엔진 적용

- Rigidboy 컴포넌트는 물리 엔진을 적용
  - 중력, 질량, 이동, 회전 등

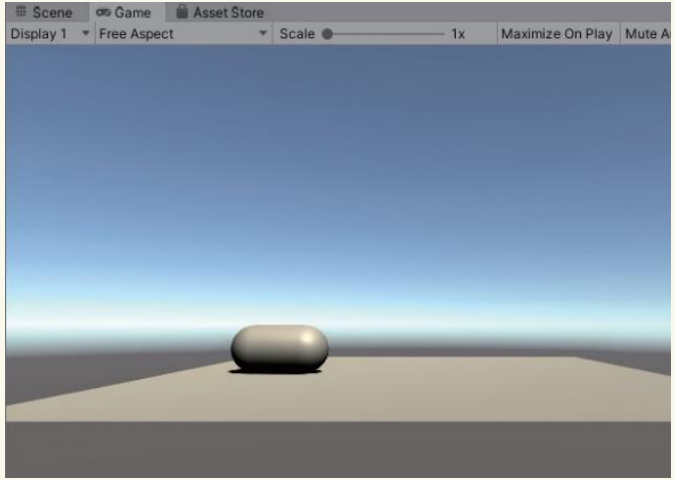


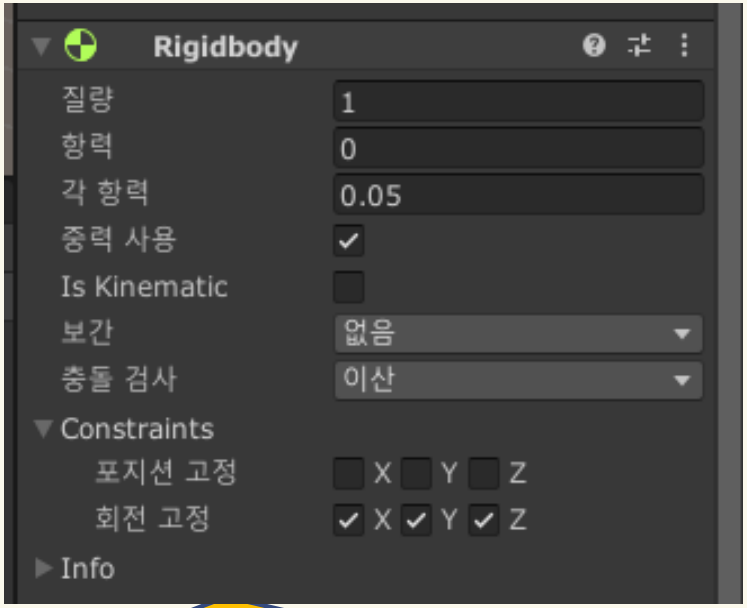
다시 실행해보면
중력의 영향으로 추락



02. 물리 엔진 적용

- 물리의 영향으로 Player를 이리저리 이동하다 보면 쓰러질 수 있음
  - Rigidbody에서 Costraints 항목의 값을 조정





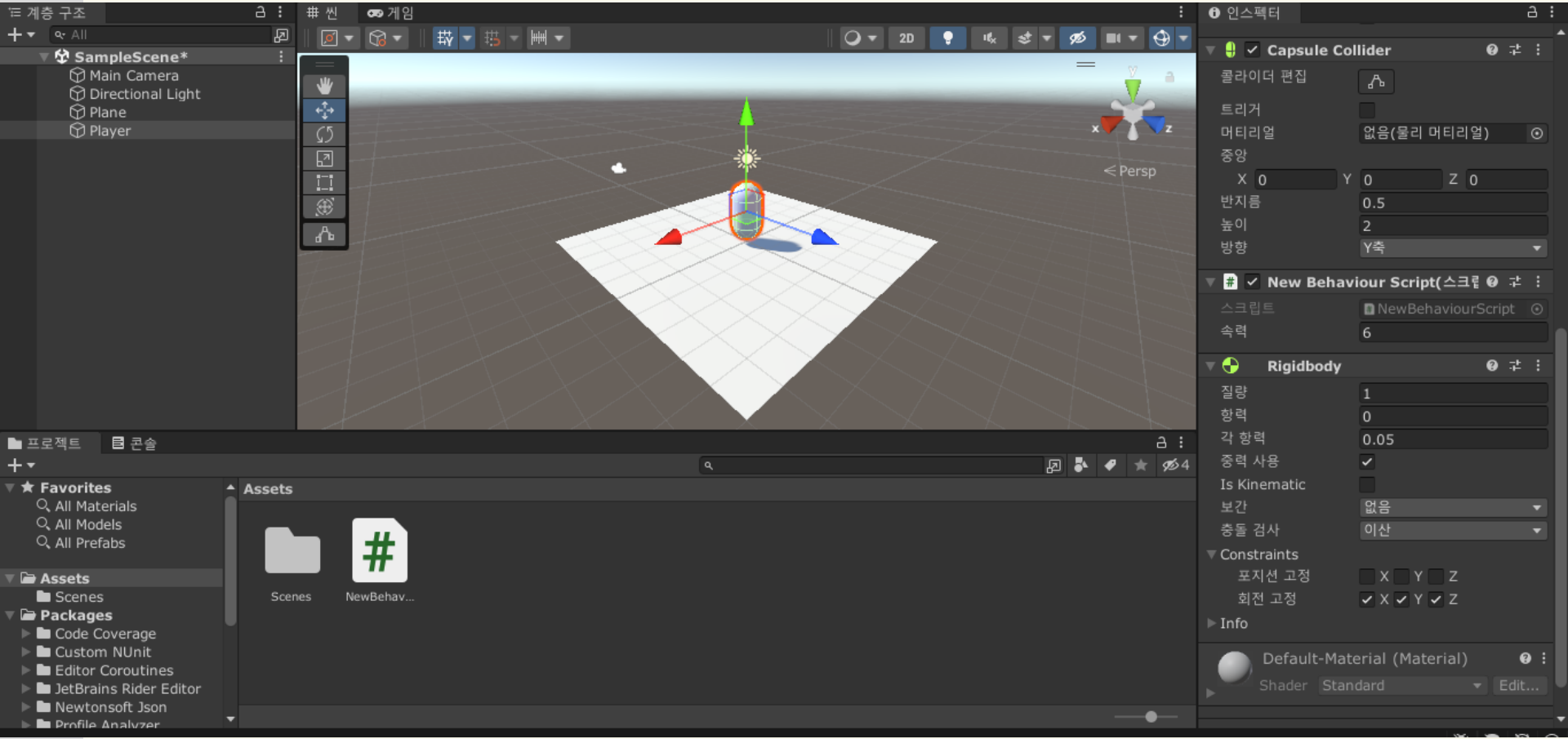
회전을 못하게 조정
Position을 체크하면 이동 못함

프리팹을 이용한 게임 제작

- 스크립트 : PlayerController
- 기본 이동 기능구현
- **좌우로 움직이는지 확인**

키코드	벡터 방향
KeyCode.RightArrow	Vector3.right
KeyCode.UpArrow	Vector3.forward
KeyCode.DownArrow	Vector3.back

```
1   using System.Collections;
2   using System.Collections.Generic;
3   using UnityEngine;
4
5   public class PlayerController : MonoBehaviour
6   {
7       float speed = 10.0f; // 이동속도
8
9       // Start is called before the first frame update
10      void Start()
11      {
12
13      }
14
15      // Update is called once per frame
16      void Update()
17      {
18          if (Input.GetKey (KeyCode.LeftArrow)) {
19              transform.Translate (Vector3.left * speed * Time.deltaTime);
20          }
21          if (Input.GetKey (KeyCode.RightArrow)) {
22              transform.Translate (Vector3.right * speed * Time.deltaTime);
23          }
24      }
25  }
```

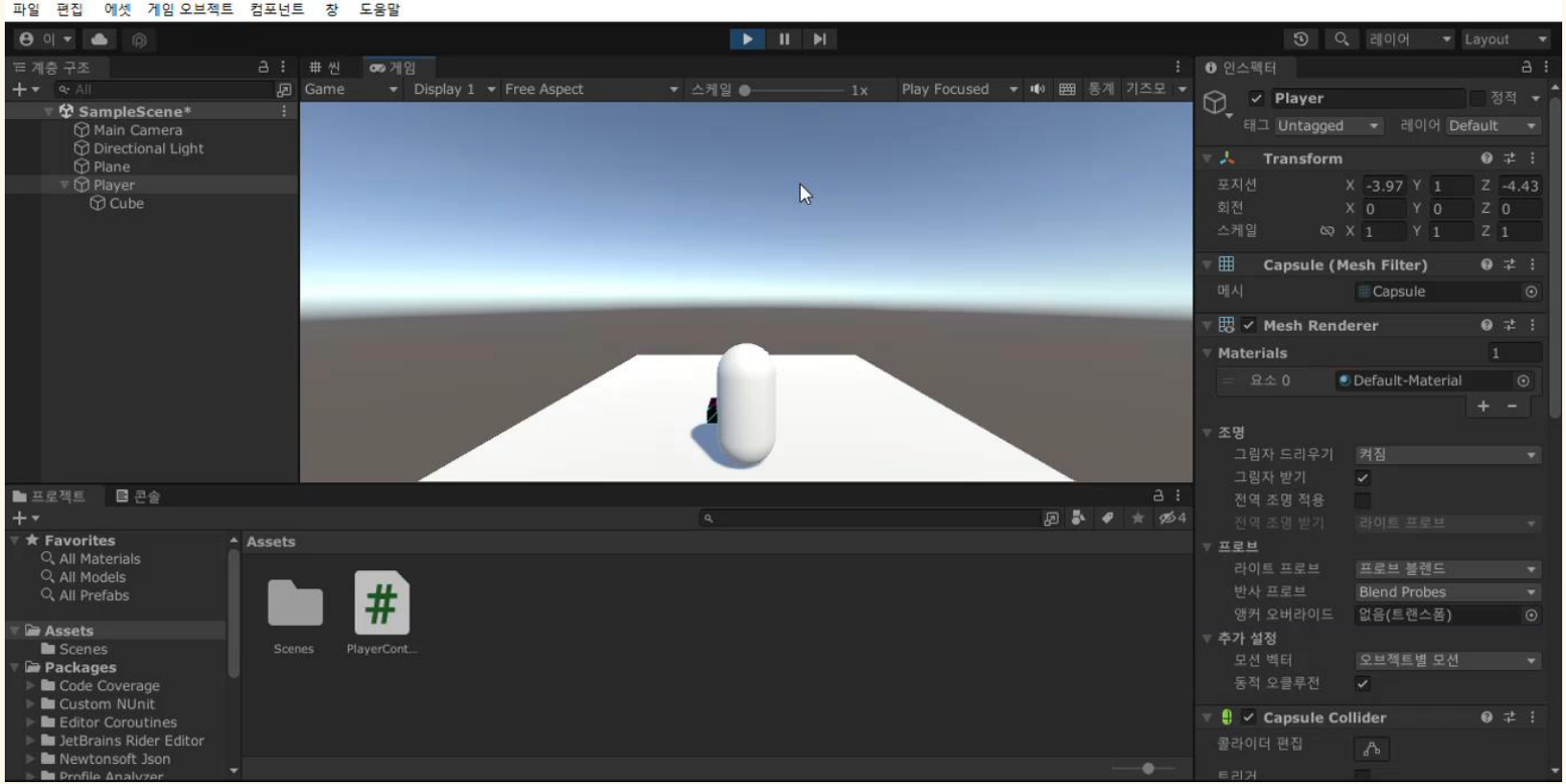

계층 구조
All
SampleScene*
Main Camera
Directional Light
Plane
Player
씬
게임
2D
Persp
인스펙터
Capsule Collider
콜라이더 편집
트리거
머티리얼
없음(물리 머티리얼)
중앙
X 0
Y 0
Z 0
반지름
0.5
높이
2
방향
Y축
New Behaviour Script(스크립
스크립트
NewBehaviourScript
속력
6
Rigidbody
질량
1
항력
0
각 항력
0.05
중력 사용
Is Kinematic
보간
없음
충돌 검사
이산
Constraints
포지션 고정
회전 고정
Info
Default-Material (Material)
Shader
Standard
Edit...
프로젝트
콘솔
Favorites
All Materials
All Models
All Prefabs
Assets
Scenes
Packages
Code Coverage
Custom NUnit
Editor Coroutines
JetBrains Rider Editor
Newtonsoft Json
Profile Analyzer
Assets
Scenes
NewBehav...

- 스크립트 :

PlayerController

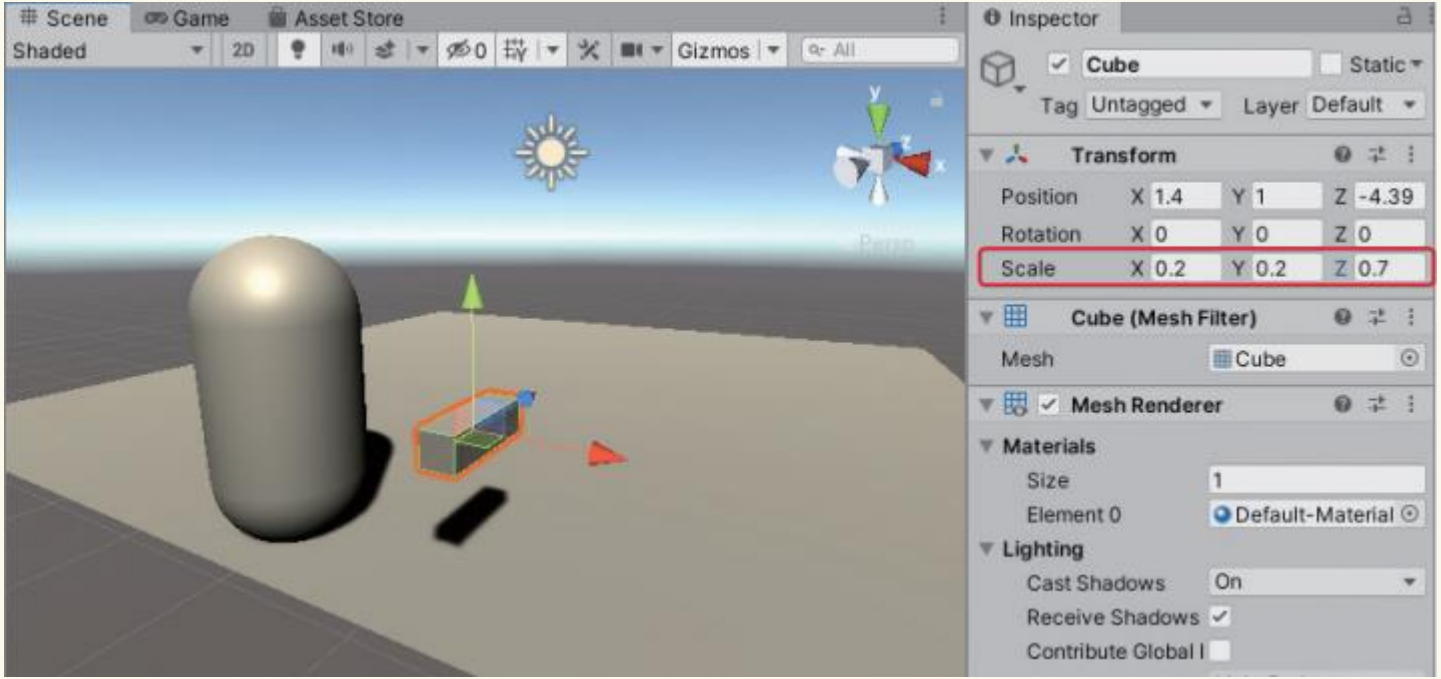
```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class PlayerController : MonoBehaviour
{
    float speed = 10.0f; // 이동속도
    // Start is called before the first frame update
    void Start()
    {
    }
    // Update is called once per frame
    void Update()
    {
        if (Input.GetKey (KeyCode.LeftArrow)) {
            transform.Translate (Vector3.left * speed * Time.deltaTime);
        }
        if (Input.GetKey (KeyCode.RightArrow)) {
            transform.Translate (Vector3.right * speed * Time.deltaTime);
        }
    }
}
```

• 동영상


파일 편집 에셋 게임 오브젝트 컴포넌트 창 도움말
계층 구조
SampleScene*
Main Camera
Directional Light
Plane
Player
Cube
씬
게임
Game
Display 1
Free Aspect
스케일
1x
Play Focused
통계
기즈모
인스펙터
Player
정적
태그 Untagged
레이어 Default
Transform
포지션 X -3.97 Y 1 Z -4.43
회전 X 0 Y 0 Z 0
스케일 X 1 Y 1 Z 1
Capsule (Mesh Filter)
메시 Capsule
Mesh Renderer
Materials 1
요소 0 Default-Material
조명
그림자 드리우기 켜짐
그림자 받기
전역 조명 적용
전역 조명 받기 라이트 프로브
프로브
라이트 프로브 프로브 블렌드
반사 프로브 Blend Probes
앵커 오버라이드 없음(트랜스폼)
추가 설정
모션 벡터 오브젝트별 모션
동적 오클루전
Capsule Collider
콜라이더 편집
프로젝트
콘솔
Favorites
All Materials
All Models
All Prefabs
Assets
Scenes
Packages
Code Coverage
Custom NUnit
Editor Coroutines
JetBrains Rider Editor
Newtonsoft Json
Assets
Scenes
PlayerCont...

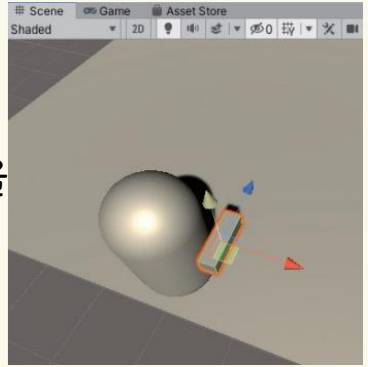
프리팹을 이용한 게임 제작

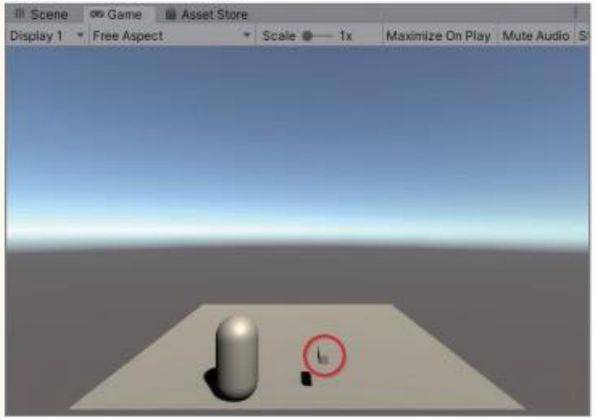
- 총 만들기 : 큐브로 간단하게 제작



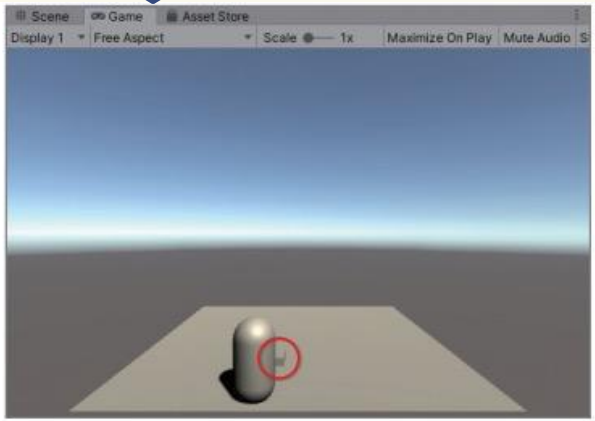
프리팹을 이용한 게임 제작

- 총의 위치 조정하기
  - Player와 같이 Gun을 움직이게 하려면 [Hierarchy] 창에서 Gun을 Player로 드래그





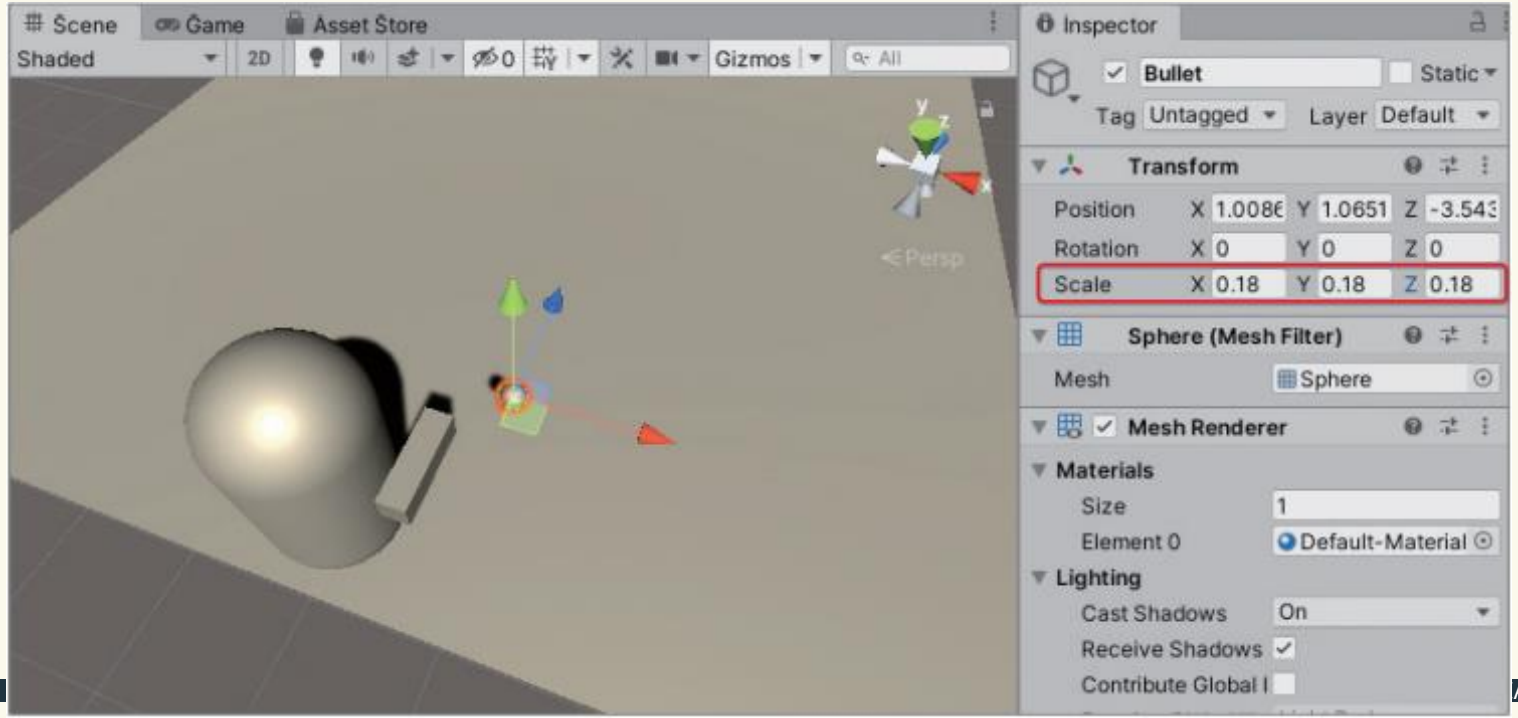
Player와 Gun이 독립적인 경우



Gun이 Player의 하위 오브젝트인 경우

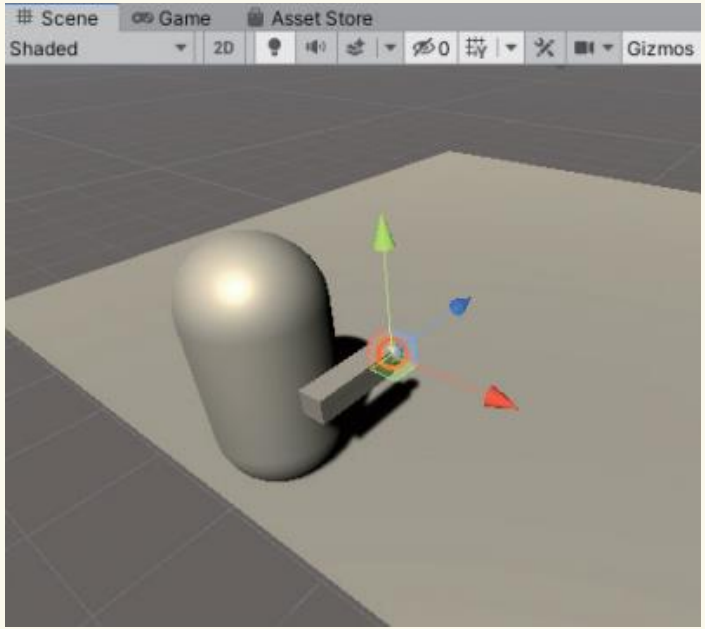
프리팹을 이용한 게임 제작

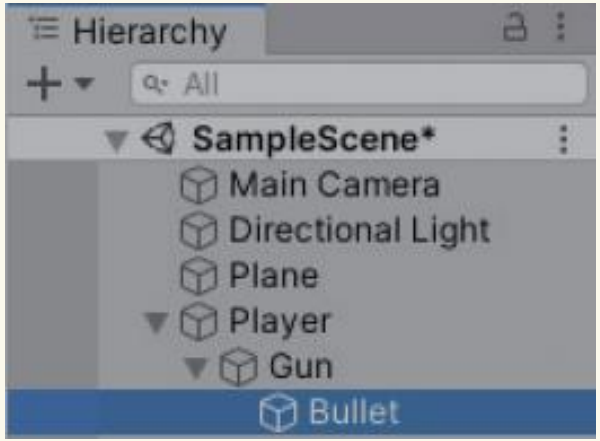
- 총알 만들어 발사하기
  - 스피어 오브젝트로 총알을 만들고 Scale의 X, Y, Z를 모두 0.18로 설정



프리팹을 이용한 게임 제작

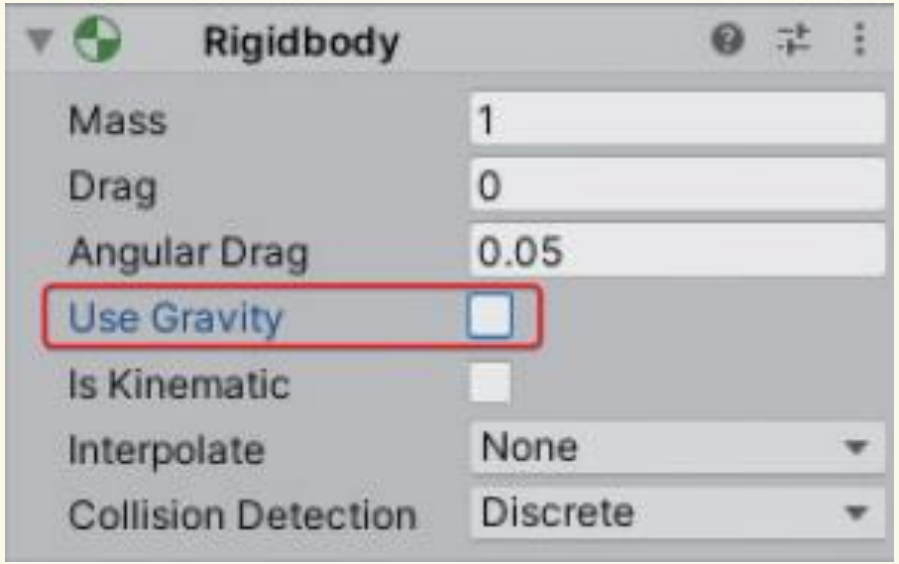
- 총알 오브젝트 위치 조정
  - 총구 앞으로 오게, Gun오브젝트의 하위로 오게 위치 조정

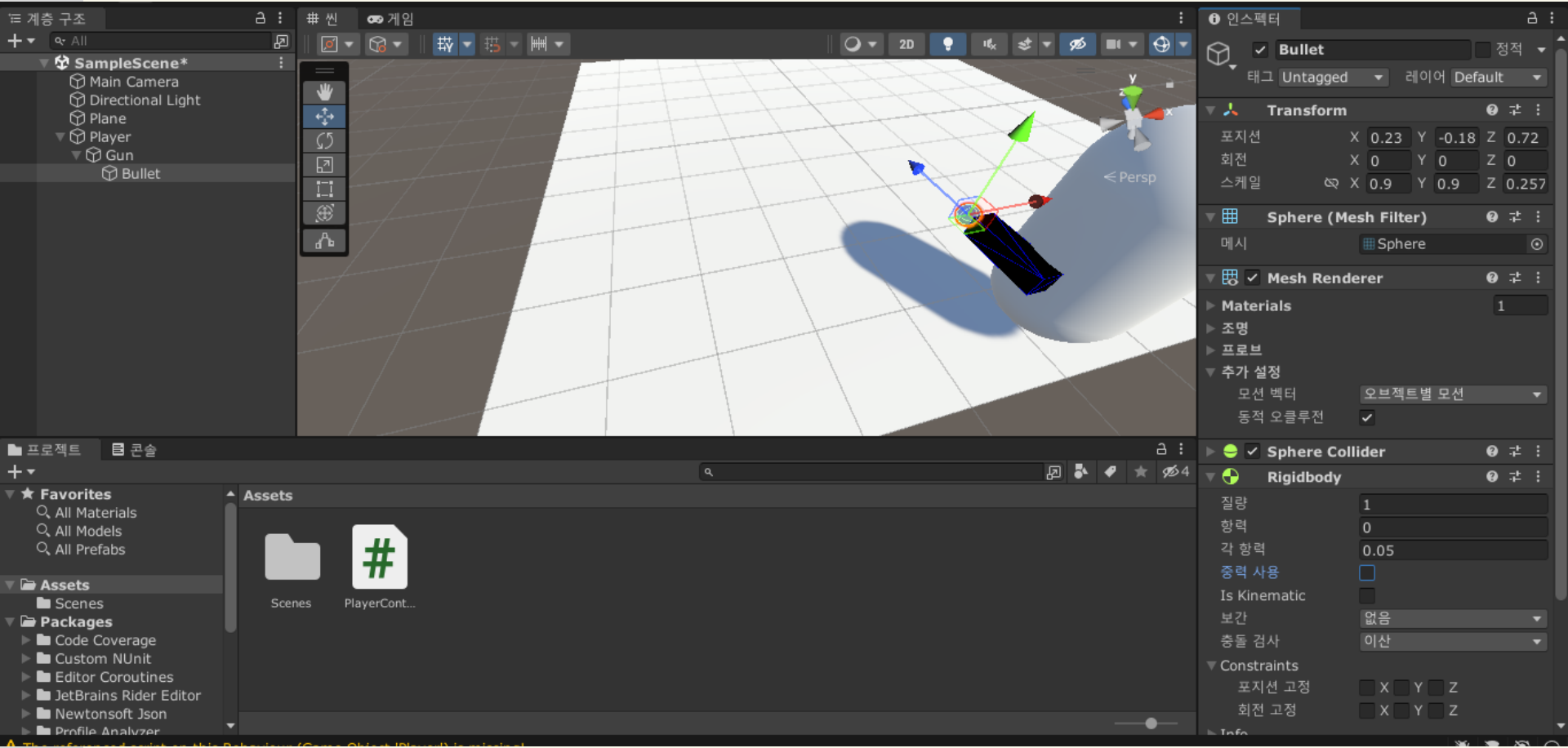




프리팹을 이용한 게임 제작

- 총알이 중력의 영향을 받지 않고 직진하도록 Rigidbody에서 중력 해제




계층 구조
All
SampleScene*
Main Camera
Directional Light
Plane
Player
Gun
Bullet
씬
게임
2D
Persp
인스펙터
Bullet
정적
태그 Untagged
레이어 Default
Transform
포지션 X 0.23 Y -0.18 Z 0.72
회전 X 0 Y 0 Z 0
스케일 X 0.9 Y 0.9 Z 0.257
Sphere (Mesh Filter)
메시 Sphere
Mesh Renderer
Materials 1
조명
프로브
추가 설정
모션 벡터 오브젝트별 모션
동적 오클루전
Sphere Collider
Rigidbody
질량 1
항력 0
각 항력 0.05
중력 사용
Is Kinematic
보간 없음
충돌 검사 이산
Constraints
포지션 고정 X Y Z
회전 고정 X Y Z
프로젝트
콘솔
Favorites
All Materials
All Models
All Prefabs
Assets
Scenes
Packages
Code Coverage
Custom NUnit
Editor Coroutines
JetBrains Rider Editor
Newtonsoft Json
Assets
Scenes
PlayerCont...

- 총알 발사 스크립트 작성
  - 마우스 왼쪽 버튼을 클릭하면 총알 발사하도록 코딩

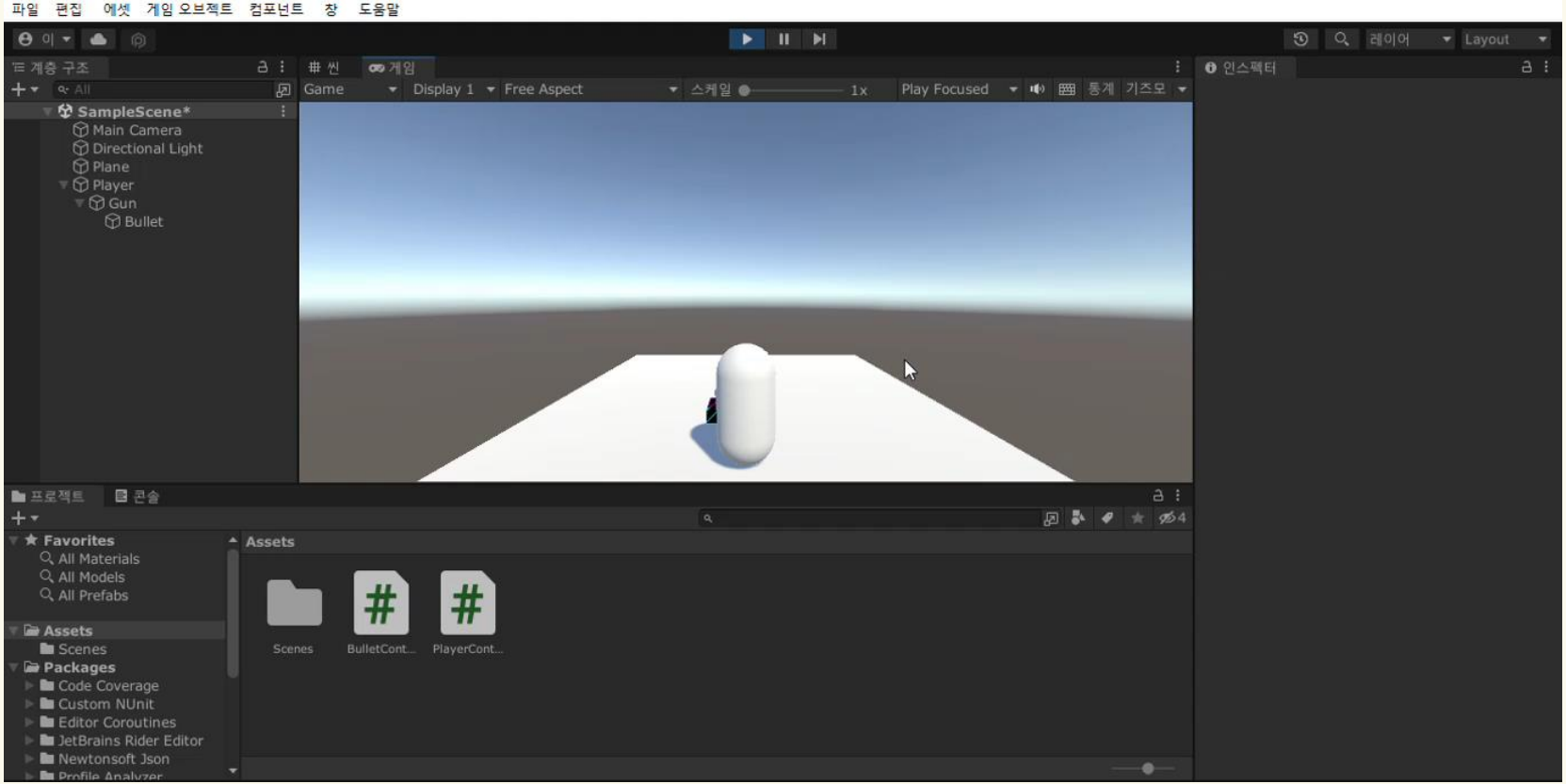
```
1   using System.Collections;
2   using System.Collections.Generic;
3   using UnityEngine;
4
5   public class BulletController : MonoBehaviour
6   {
7       // Start is called before the first frame update
8       void Start()
9       {
10
11      }
12      // Update is called once per frame
13      void Update()
14      {
15          if (Input.GetMouseButtonDown (0)) {  // 마우스 좌클릭 인식
16              Shoot (new Vector3 (0 , 0 , 50)); // Shoot() 함수 호출
17          }
18      }
19      // 총알 발사 함수
20      public void Shoot (Vector3 dir)
21      {
22          GetComponent<Rigidbody> ().AddForce (dir);
23      }
24  }
25
```

- **스크립트 :**

BulletController

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
public class BulletController : MonoBehaviour
{
    // Start is called before the first frame update
    void Start()
    {
    }
    // Update is called once per frame
    void Update()
    {
        if (Input.GetMouseButtonDown (0)) {  // 마우스 좌클릭 인식
            Shoot (new Vector3 (0 , 0 , 50)); // Shoot() 함수 호출
        }
    }
    // 총알 발사 함수
    public void Shoot (Vector3 dir)
    {
        GetComponent<Rigidbody> ().AddForce (dir);
    }
}
```

• 동영상


파일 편집 에셋 게임 오브젝트 컴포넌트 창 도움말
계층 구조
씬
게임
인스펙터
Game
Display 1
Free Aspect
스케일
1x
Play Focused
통계
기즈모
레이어
Layout
SampleScene*
Main Camera
Directional Light
Plane
Player
Gun
Bullet
프로젝트
콘솔
Favorites
All Materials
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All Prefabs
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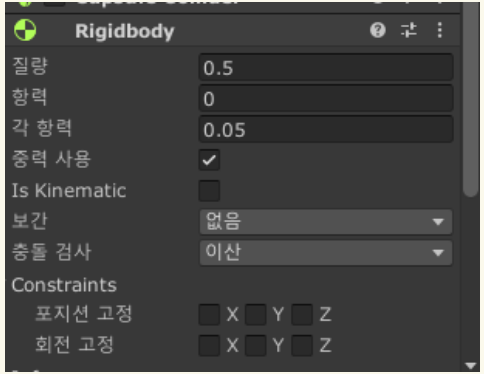
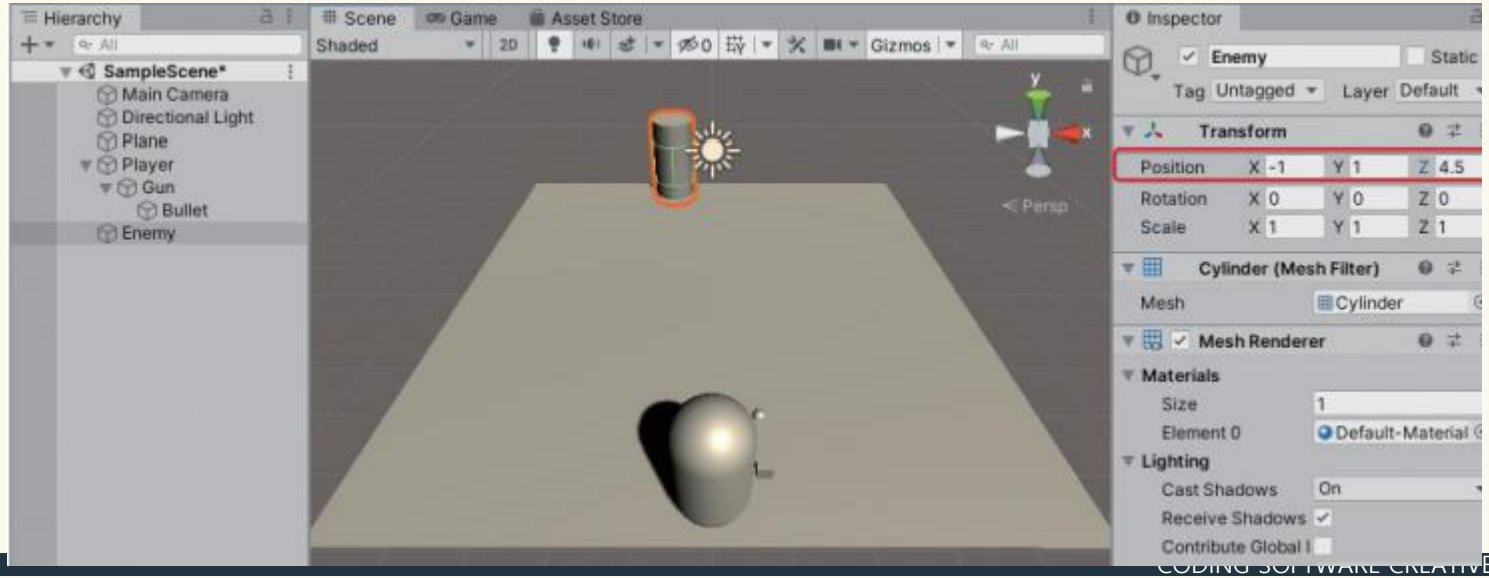
Scenes
Packages
Code Coverage
Custom NUnit
Editor Coroutines
JetBrains Rider Editor
Newtonsoft Json
Assets
Scenes
BulletCont...
PlayerCont...

프리팹을 이용한 게임 제작

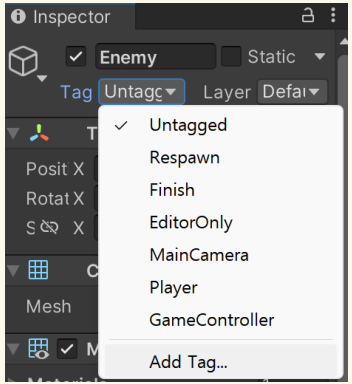
- 충돌처리하기
  - 적 캐릭터 배치(실린더 오브젝트)
  - 태그 생성(ENEMY)[Tag-Add Tag-ENEMY 추가]
  - 총알이 적과 충돌하면 총알 제거

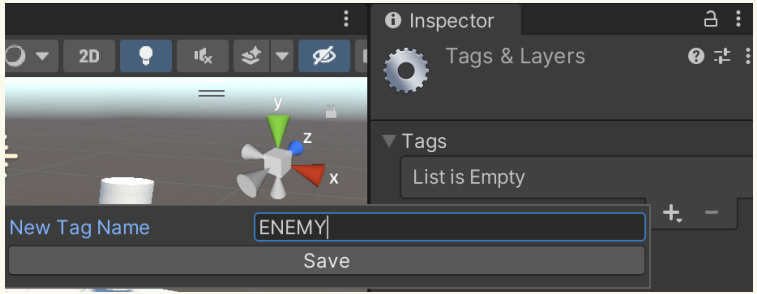


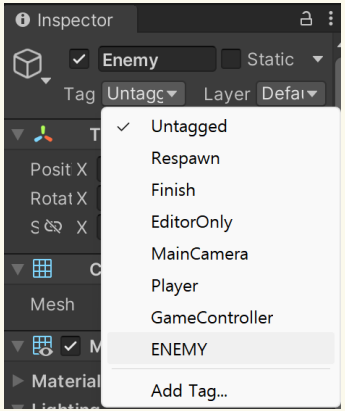


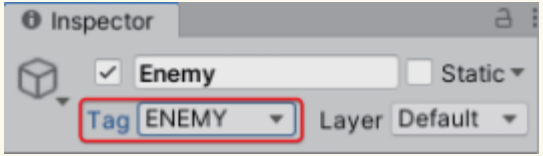
프리팹을 이용한 게임 제작

- 충돌처리하기
  - 태그 생성(ENEMY)[Tag-Add Tag-ENEMY 추가]





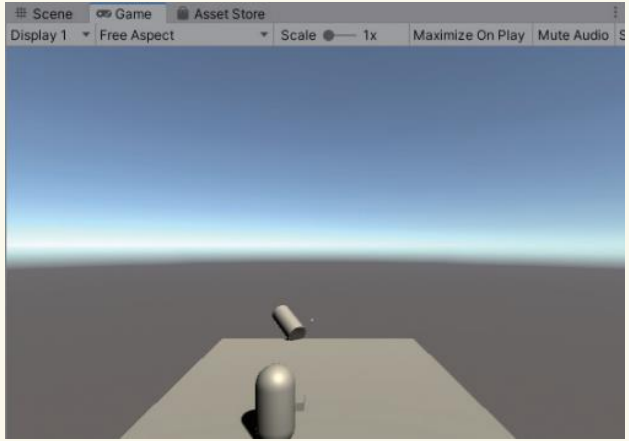




프리팹을 이용한 게임 제작

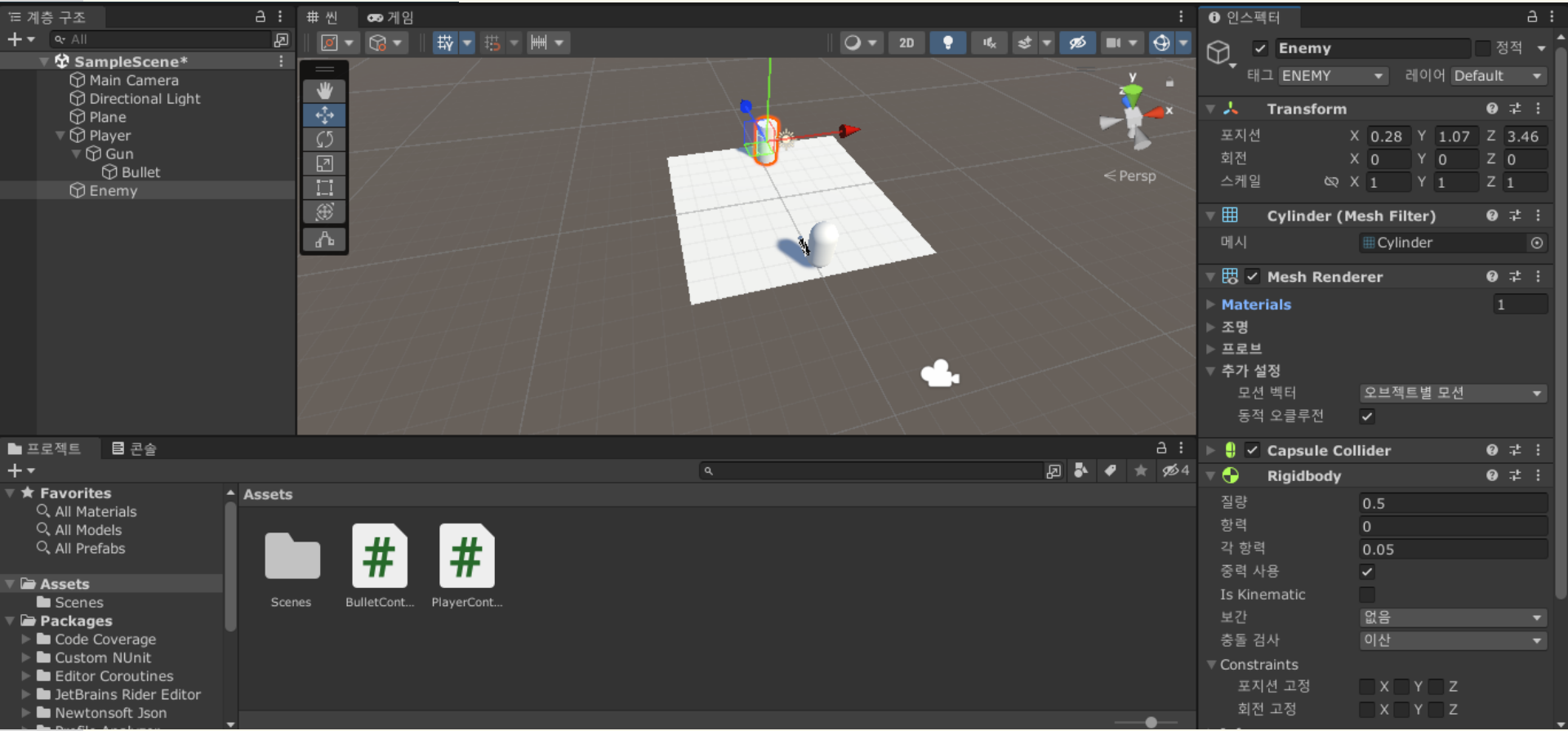
- 총알이 적과 충돌하면 총알 제거

BulletController 아래에 추가




```
void OnCollisionEnter (Collision coll)
{
    if (coll.collider.tag == "ENEMY")
    {
        Destroy (gameObject, 0.2f);   // 오브젝트 파괴
    }
}
```

충돌하고 0.2초 후에
오브젝트 삭제


계층 구조
All
SampleScene*
Main Camera
Directional Light
Plane
Player
Gun
Bullet
Enemy
씬
게임
2D
Persp
인스펙터
Enemy
정적
태그 ENEMY
레이어 Default
Transform
포지션 X 0.28 Y 1.07 Z 3.46
회전 X 0 Y 0 Z 0
스케일 X 1 Y 1 Z 1
Cylinder (Mesh Filter)
메시 Cylinder
Mesh Renderer
Materials 1
조명
프로브
추가 설정
모션 벡터 오브젝트별 모션
동적 오클루전
Capsule Collider
Rigidbody
질량 0.5
항력 0
각 항력 0.05
중력 사용
Is Kinematic
보간 없음
충돌 검사 이산
Constraints
포지션 고정 X Y Z
회전 고정 X Y Z
프로젝트
콘솔
Favorites
All Materials
All Models
All Prefabs
Assets
Scenes
Packages
Code Coverage
Custom NUnit
Editor Coroutines
JetBrains Rider Editor
Newtonsoft Json
Assets
Scenes
BulletCont...
PlayerCont...

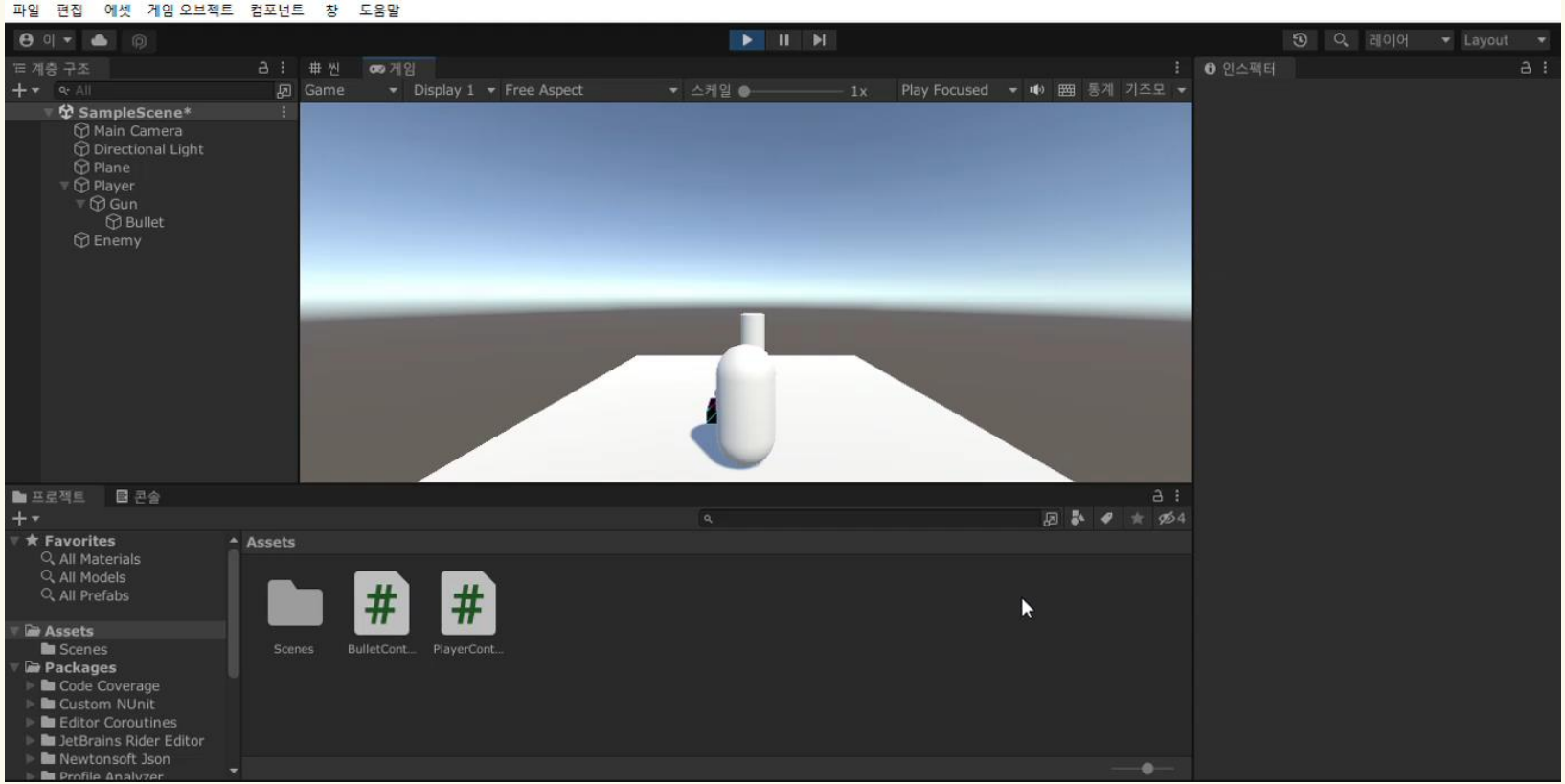
- 스크립트 :

BulletController

추가

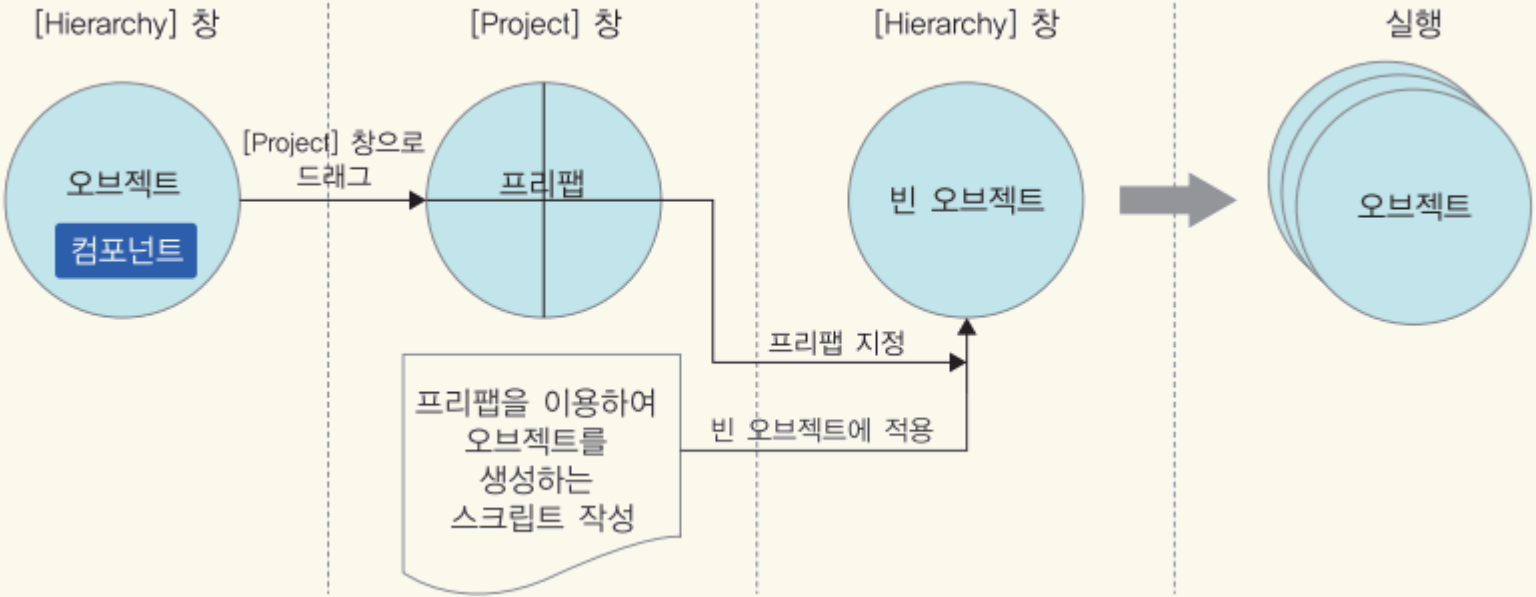
```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
public class BulletController : MonoBehaviour
{
    void Start()
    {
    }
    // Update is called once per frame
    void Update()
    {
        if (Input.GetMouseButtonDown (0)) {  // 마우스 좌클릭 인식
           Shoot (new Vector3 (0 , 0 , 50)); // Shoot() 함수 호출
        }
    }
    // 총알 발사 함수
    public void Shoot (Vector3 dir)
    {
        GetComponent<Rigidbody> ().AddForce (dir);
    }
    void OnCollisionEnter (Collision coll)
    {
        if (coll.collider.tag == "ENEMY") {
            Destroy (gameObject, 0.2f);   // 오브젝트 파괴
        }
    }
}
```

• 동영상

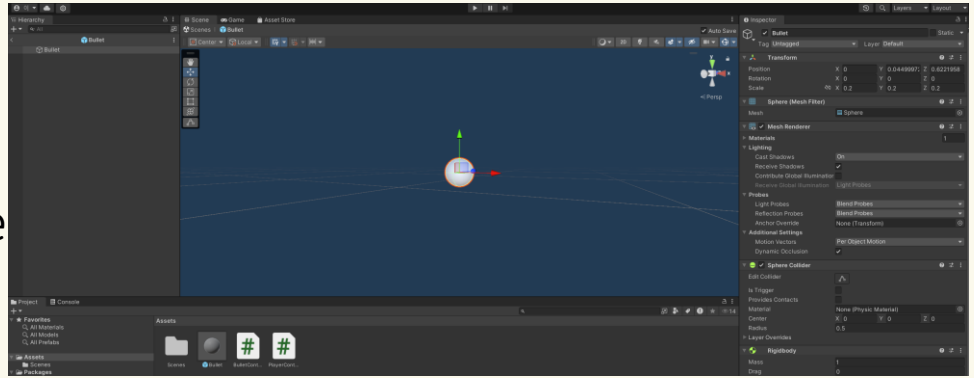

파일 편집 에셋 게임 오브젝트 컴포넌트 창 도움말
계층 구조
씬
게임
인스펙터
Game
Display 1
Free Aspect
스케일
1x
Play Focused
통계
기즈모
레이어
Layout
SampleScene*
Main Camera
Directional Light
Plane
Player
Gun
Bullet
Enemy
프로젝트
콘솔
Favorites
All Materials
All Models
All Prefabs
Assets
Scenes
Packages
Code Coverage
Custom NUnit
Editor Coroutines
JetBrains Rider Editor
Newtonsoft Json
Assets
Scenes
BulletCont...
PlayerCont...

프리팹을 이용한 게임 제작

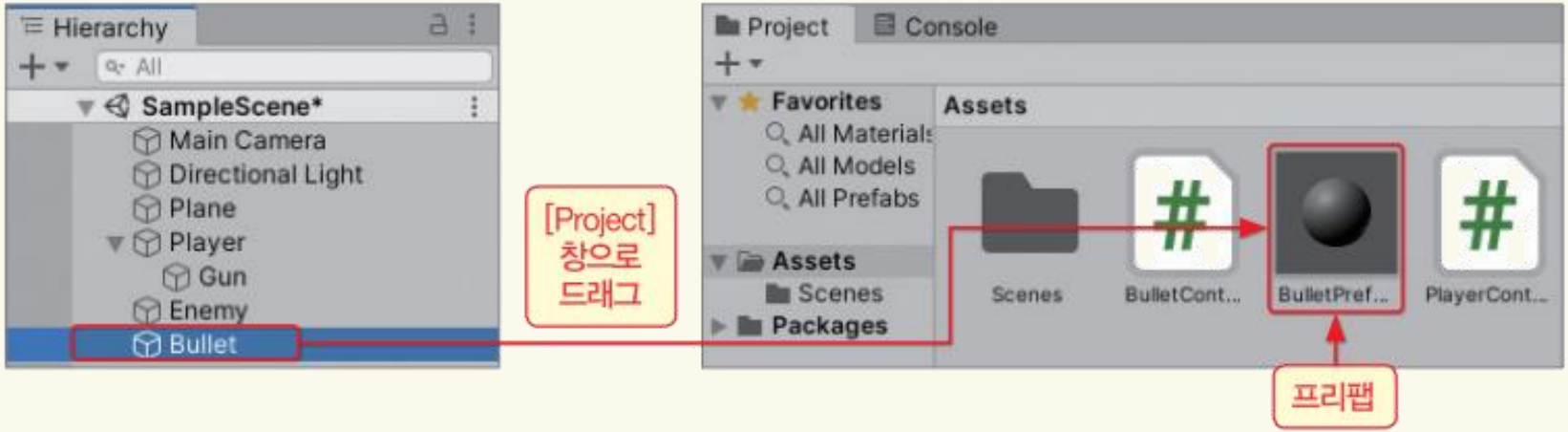
- 프리팹이란?
  - 동일한 오브젝트를 같은 형태로 계속 생성할 때 사용
  - 프리팹은 일종의 설계도



프리팹을 이용한 게임 제작



- 총알을 프리팹으로 만들기 위해 [Hie
  창으로 드래그
  - 프리팹의 이름을 bulletPrefab으로 수정



프리팹을 이용한 게임 제작

- 스크립트 작성하기: BulletGenerator

```
public GameObject bulletPrefab;
```

프리팹 정보를 지정할 변수

```
Instantiate(GameObject original , Vector3 position , Quaternion rotation)
```

프리팹으로 오브젝트를 생성하는 함수

프리팹을 이용

```
public class BulletGenerator : MonoBehaviour
{
    public GameObject bulletPrefab;   // 프리팹을 지정하는 변수

    // Start is called before the first frame update
    void Start()
    {

    }

    // Update is called once per frame
    void Update()
    {
        if (Input.GetMouseButtonDown (0))
        {
            // 프리팹을 이용하여 오브젝트 생성
            GameObject bullet = Instantiate (bulletPrefab , transform.position ,
                                             transform.rotation);

            // BulletController 스크립트를 찾아서 Shoot() 함수 호출(총알 발사)
            bullet.GetComponent<BulletController> ().Shoot (new Vector3 (0 , 0 , 100));
        }
    }
}
```

프리팹 정보를 지정할 변수 선언

프리팹으로 총알 생성

총알 발사

- 스크립트 : **BulletGenerator**

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class BulletGenerator : MonoBehaviour
{
    public GameObject bulletPrefab;   // Prefab을 지정할 변수
    // Start is called before the first frame update
    void Start()
    {
    }
    // Update is called once per frame
    void Update ()
    {
        if (Input.GetMouseButtonDown (0)) {
            // Prefab을 이용하여 오브젝트 생성
            GameObject bullet = Instantiate (bulletPrefab , transform.position , transform.rotation);
            // BulletController 스크립트를 찾아서 Shoot() 함수 호출(총알 발사)
            bullet.GetComponent<BulletController> ().Shoot (new Vector3 (0 , 0 , 50));
        }
    }
}
```

프리팹을 이용한 게임 제작

- BulletController 스크립트의 Update( )에 작성한 코드를 삭제하거나 주석 처리

```
void Update()
{
    //   if (Input.GetMouseButtonDown (0))   // 마우스 왼쪽 버튼 클릭 인식
    //   {
    //      Shoot (new Vector3 (0 , 0 , 50)); // Shoot() 함수 호출
    //   }
}
```

- 스크립트 :

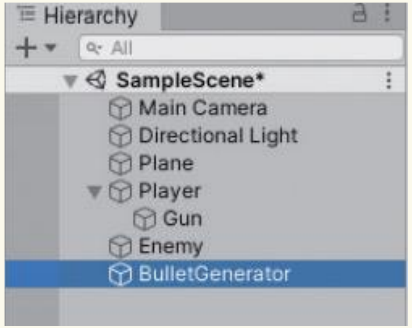
BulletController

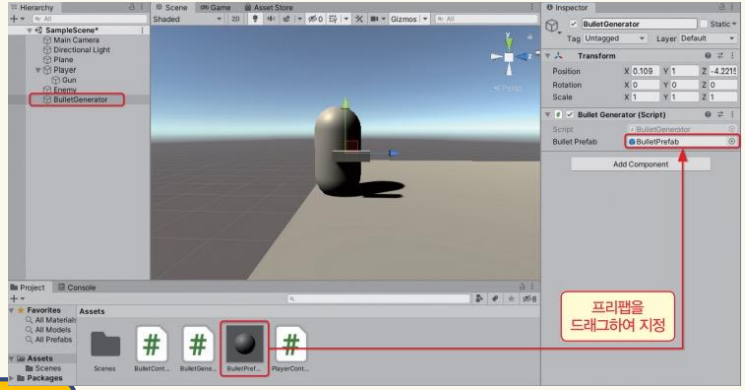
수정[삭제]

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
public class BulletController : MonoBehaviour
{
    void Start()
    {
    }
    // Update is called once per frame
    void Update()
    {
        //if (Input.GetMouseButtonDown (0)) {  // 마우스 좌클릭 인식
        //    Shoot (new Vector3 (0 , 0 , 50)); // Shoot() 함수 호출
        // }
    }
    // 총알 발사 함수
    public void Shoot (Vector3 dir)
    {
        GetComponent<Rigidbody> ().AddForce (dir);
    }
    void OnCollisionEnter (Collision coll)
    {
        if (coll.collider.tag == "ENEMY") {
            Destroy (gameObject, 0.2f);   // 오브젝트 파괴
        }
    }
}
```

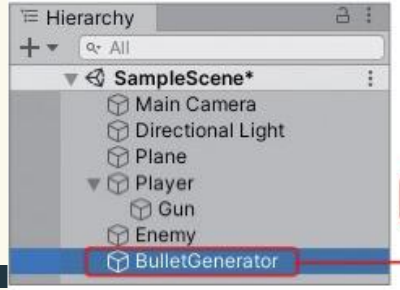
프리팹을 이용한 게임 제작

- BulletGenerator 스크립트를 적용할 빈 오브젝트 추가: BulletGenerator

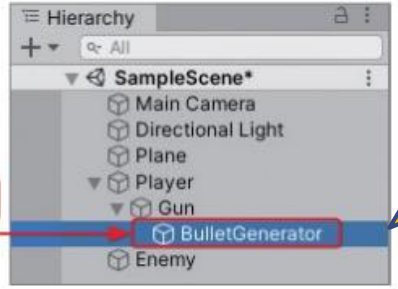




BulletGenerator 생성 및
스크립트 적용

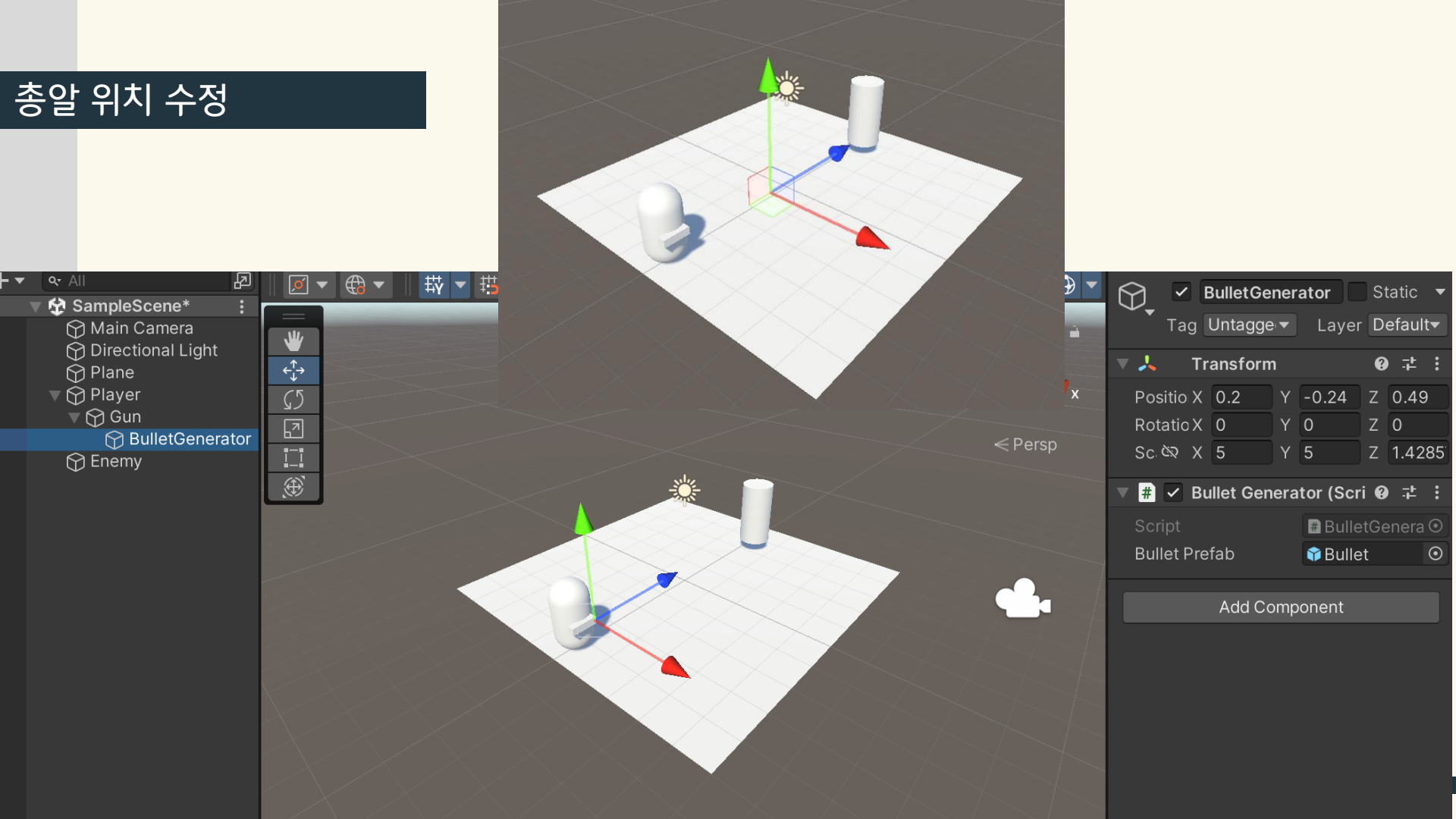


위치 변경



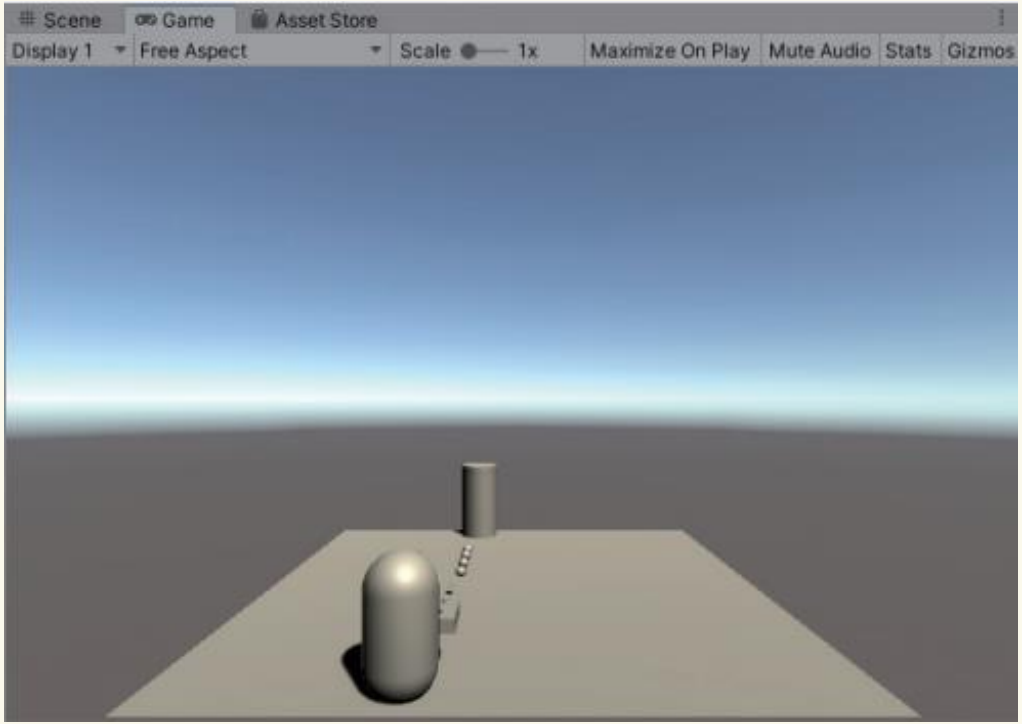
BulletGenerator
위치 조정

총알 위치 수정


All
SampleScene*
Main Camera
Directional Light
Plane
Player
Gun
BulletGenerator
Enemy
Persp
BulletGenerator
Static
Tag Untagge
Layer Default
Transform
Positio X 0.2 Y -0.24 Z 0.49
Rotatio X 0 Y 0 Z 0
Sc X 5 Y 5 Z 1.4285
Bullet Generator (Scri
Script BulletGenera
Bullet Prefab Bullet
Add Component

프리팹을 이용한 게임 제작

- 실행하여 확인



File Edit Assets GameObject Component Window Help

Hierarchy

- SampleScene*
  - Main Camera
  - Directional Light
  - Plane
  - Player
    - Gun
      - BulletGenerator
  - Enemy

Scene | Game

Game | Display 1 | Free Aspect | Scale 2x | Play Focused | Stats | Gizmos

Inspector

Main Camera — Static

Tag MainCamera — Layer Default

Transform

Position	X 1.69	Y 3.66	Z -11.58
Rotation	X 0	Y 0	Z 0
Scale	X 1	Y 1	Z 1

Camera

Clear Flags	Skybox
Background	
Culling Mask	Everything
Projection	Perspective
FOV Axis	Vertical
Field of View	60
Physical Camera	
Clipping Planes	Near 0.3, Far 1000
Viewport Rect	X 0, Y 0, W 1, H 1
Depth	-1
Rendering Path	Use Graphics Settings
Target Texture	None (Render Texture)
Occlusion Culling	✓
HDR	Use Graphics Settings
MSAA	Use Graphics Settings
Allow Dynamic Resol	
Target Display	Display 1

Audio Listener

Add Component

Project | Console

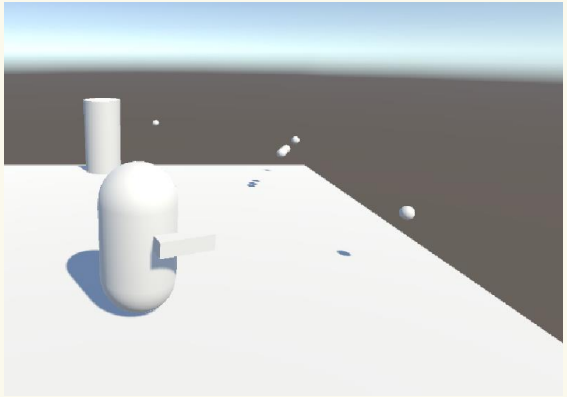
- Favorites
  - All Materials
  - All Models
  - All Prefabs
- Assets
  - Scenes
- Packages
  - Code Coverage
  - Custom NUnit
  - Editor Coroutines
  - JetBrains Rider Editor
  - Profile Analyzer
  - Settings Manager
  - Test Framework
  - TextMeshPro
  - Timeline

Assets

Scenes | Bullet | BulletCont... | BulletGene... | PlayerCont...

ENTER

캐릭터 회전하기



- 총알이 캐릭터 회전방향에서 발사될 수 있도록 수정
  - **스크립트 : BulletGenerator**

```
bullet.GetComponent<BulletController> ().Shoot (new Vector3 (0 , 0 , 50));
```

➔ Vector3 shootDirection = transform.forward;

➔ bullet.GetComponent<BulletController>().Shoot(shootDirection * 50);

```
if (Input.GetMouseButtonDown (0)) {
```

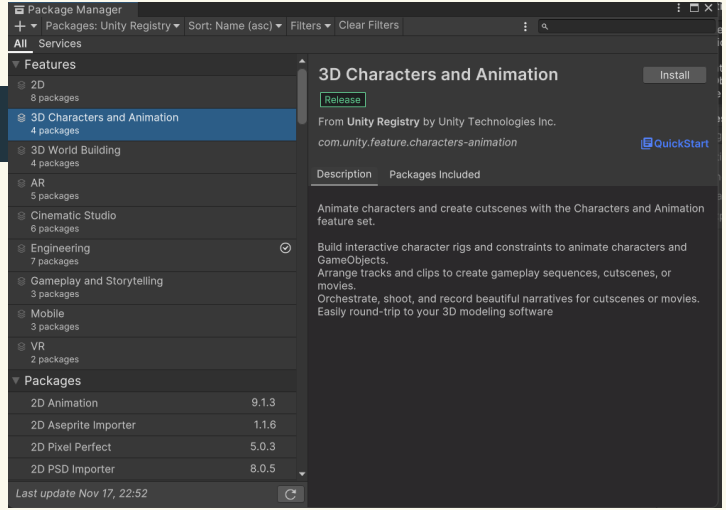
➔ if (Input.GetMouseButtonDown (1)) { //마우스 오른쪽 버튼 클릭

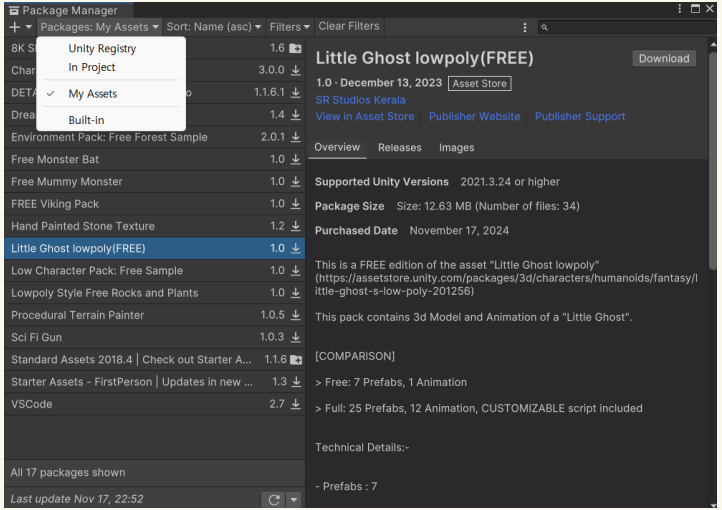
캐릭터 회전하기

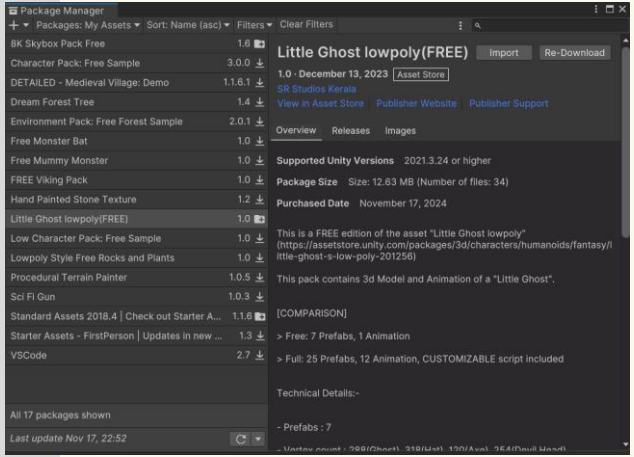
- 캐릭터 4개 방향 이동 및 회전하기[코드 추가]
  - **스크립트** : PlayerController.cs

```
if(Input.GetKey(KeyCode.UpArrow) == true){
    transform.Translate( Vector3.forward * speed * Time.deltaTime);
}
if(Input.GetKey(KeyCode.DownArrow) == true){
    transform.Translate( Vector3.back * speed * Time.deltaTime);
}
if (Input.GetMouseButtonDown(0)) { //왼쪽 마우스 클릭 회전
    transform.Rotate(0f, 1 * speed, 0f, Space.World);
}
```

에셋 가져오기


Package Manager
Packages: Unity Registry
Sort: Name (asc)
Filters
Clear Filters
All
Services
Features
2D
8 packages
3D Characters and Animation
4 packages
3D World Building
4 packages
AR
5 packages
Cinematic Studio
6 packages
Engineering
7 packages
Gameplay and Storytelling
3 packages
Mobile
3 packages
VR
2 packages
Packages
2D Animation 9.1.3
2D Aseprite Importer 1.1.6
2D Pixel Perfect 5.0.3
2D PSD Importer 8.0.5
Last update Nov 17, 22:52
3D Characters and Animation
Install
Release
From Unity Registry by Unity Technologies Inc.
com.unity.feature.characters-animation
QuickStart
Description
Packages Included
Animate characters and create cutscenes with the Characters and Animation feature set.
Build interactive character rigs and constraints to animate characters and GameObjects.
Arrange tracks and clips to create gameplay sequences, cutscenes, or movies.
Orchestrate, shoot, and record beautiful narratives for cutscenes or movies.
Easily round-trip to your 3D modeling software


Package Manager
Packages: My Assets
Sort: Name (asc)
Filters
Clear Filters
Unity Registry
In Project
My Assets
Built-in
Environment Pack: Free Forest Sample 2.0.1
Free Monster Bat 1.0
Free Mummy Monster 1.0
FREE Viking Pack 1.0
Hand Painted Stone Texture 1.2
Little Ghost lowpoly(FREE) 1.0
Low Character Pack: Free Sample 1.0
Lowpoly Style Free Rocks and Plants 1.0
Procedural Terrain Painter 1.0.5
Sci Fi Gun 1.0.3
Standard Assets 2018.4 | Check out Starter A... 1.1.6
Starter Assets - FirstPerson | Updates in new ... 1.3
VSCode 2.7
All 17 packages shown
Last update Nov 17, 22:52
Little Ghost lowpoly(FREE)
Download
1.0 · December 13, 2023
Asset Store
SR Studios Kerala
View in Asset Store
Publisher Website
Publisher Support
Overview
Releases
Images
Supported Unity Versions 2021.3.24 or higher
Package Size Size: 12.63 MB (Number of files: 34)
Purchased Date November 17, 2024
This is a FREE edition of the asset "Little Ghost lowpoly" (https://assetstore.unity.com/packages/3d/characters/humanoids/fantasy/little-ghost-s-low-poly-201256)
This pack contains 3d Model and Animation of a "Little Ghost".
[COMPARISON]
> Free: 7 Prefabs, 1 Animation
> Full: 25 Prefabs, 12 Animation, CUSTOMIZABLE script included
Technical Details:-
- Prefabs : 7


Package Manager
Packages: My Assets
Sort: Name (asc)
Filters
Clear Filters
Little Ghost lowpoly(FREE)
Import
Re-Download
1.0 · December 13, 2023
Asset Store
SR Studios Kerala
View in Asset Store
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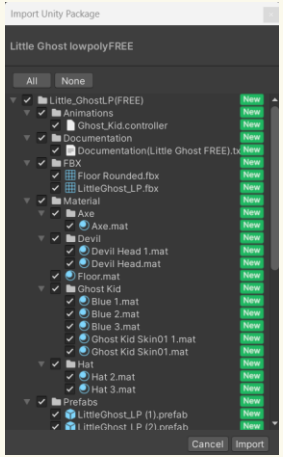
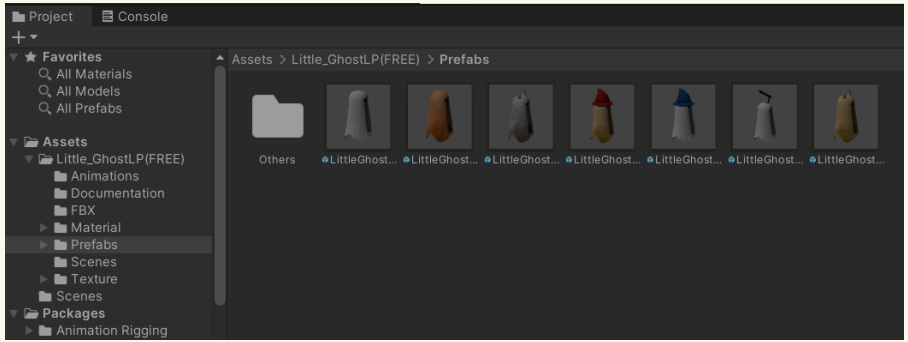
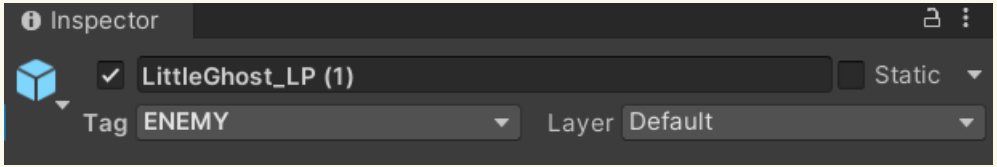
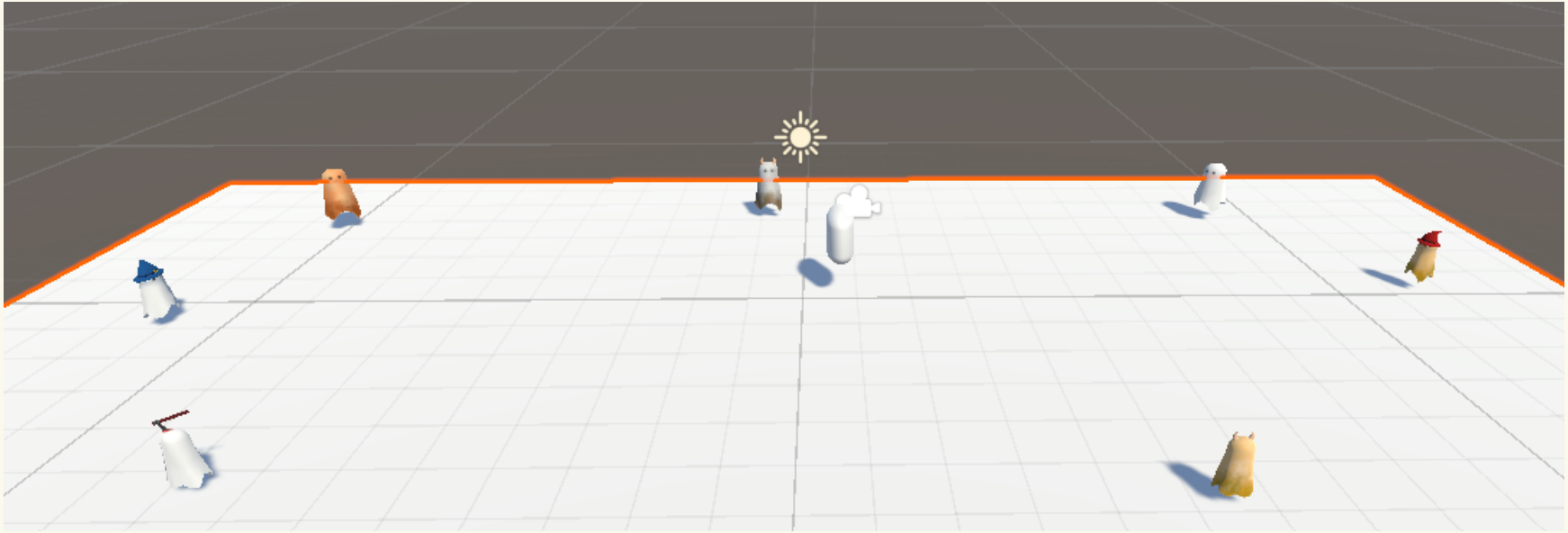
All 17 packages shown
Last update Nov 17, 22:52


Import Unity Package
Little Ghost lowpolyFREE
All
None
Little_GhostLP(FREE)
Animations
Ghost_Kid.controller
Documentation
Documentation(Little Ghost FREE).txt
FBX
Floor Rounded.fbx
LittleGhost_LP.fbx
Material
Axe
Axe.mat
Devil
Devil Head 1.mat
Devil Head.mat
Floor.mat
Ghost Kid
Blue 1.mat
Blue 2.mat
Blue 3.mat
Ghost Kid Skin01 1.mat
Ghost Kid Skin01.mat
Hat
Hat 2.mat
Hat 3.mat
Prefabs
LittleGhost_LP (1).prefab
New
Cancel
Import

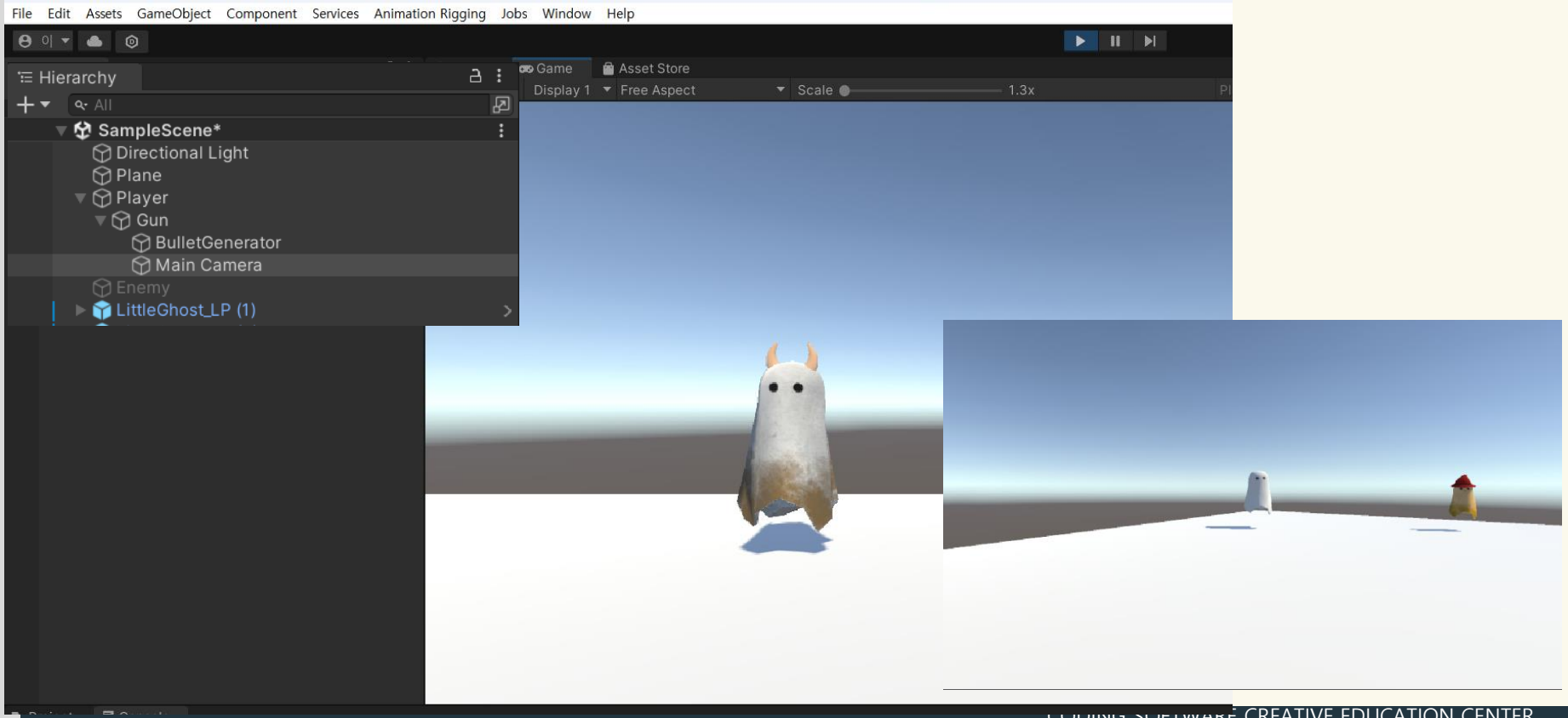

Project
Console
Favorites
All Materials
All Models
All Prefabs
Assets
Little_GhostLP(FREE)
Animations
Documentation
FBX
Material
Prefabs
Scenes
Texture
Scenes
Packages
Animation Rigging
Assets > Little_GhostLP(FREE) > Prefabs
Others
LittleGhost...

에셋 사용하기



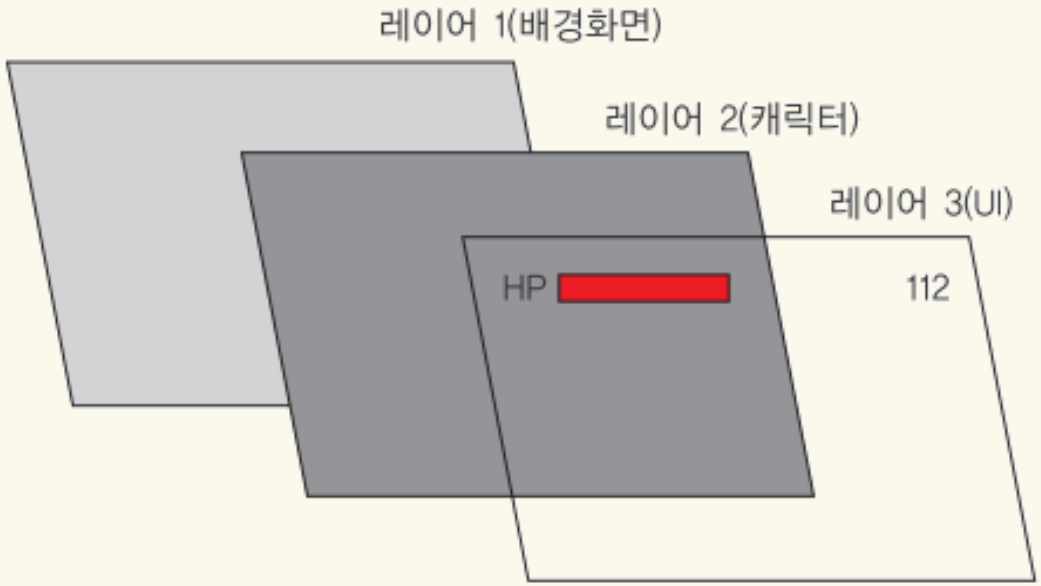


1인칭 시점으로 동작하기



프리팹을 이용한 게임 제작

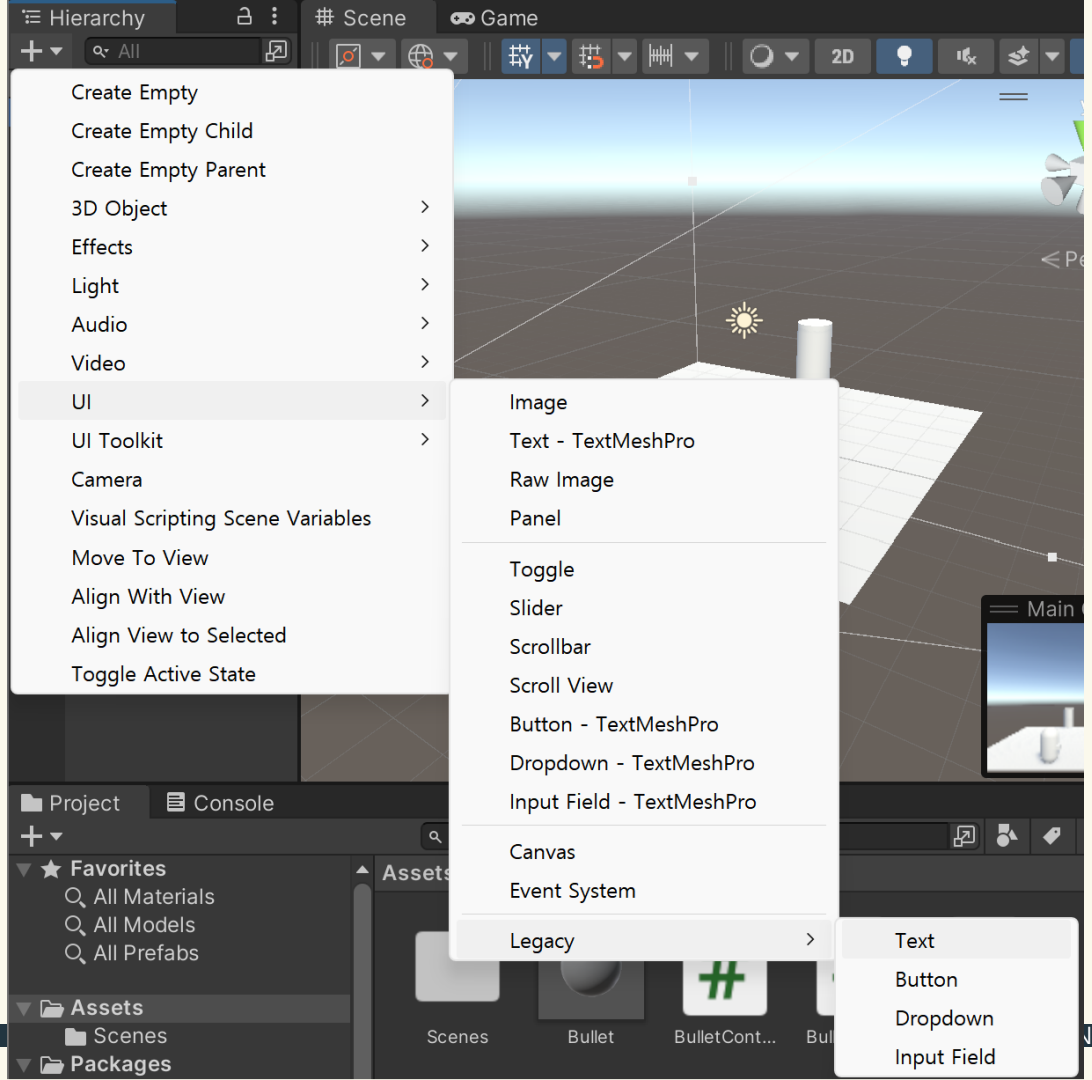
- 레이어란?
  - 게임이 플레이되는 화면과 UI가 나타나는 화면은 서로 다른 레이어



게임 레이어의 개념

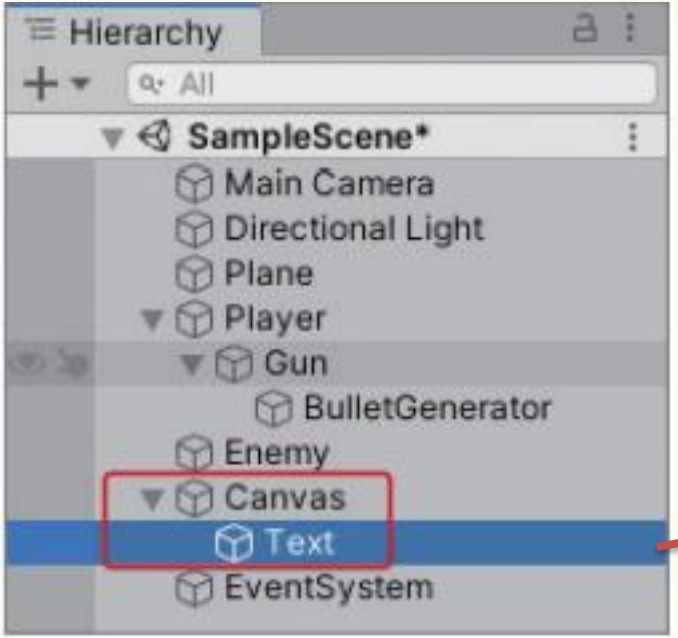
프리팹을 이용한 게임 제작

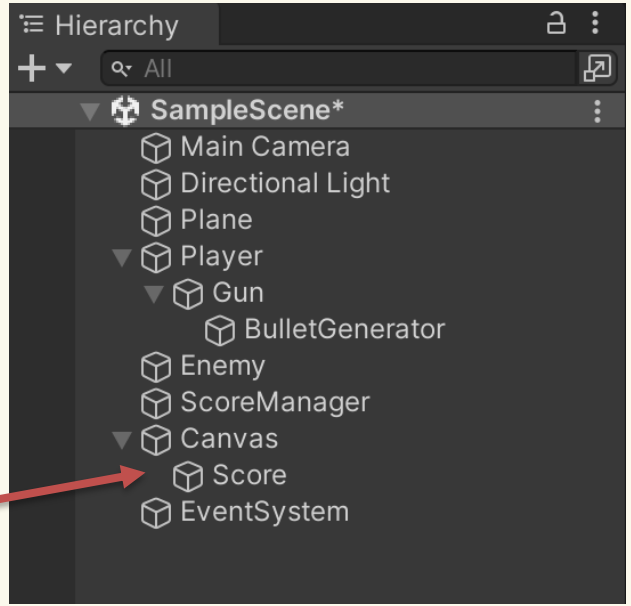
- UI 오브젝트 배치
- UI-Legacy-Text



프리팹을 이용한 게임 제작

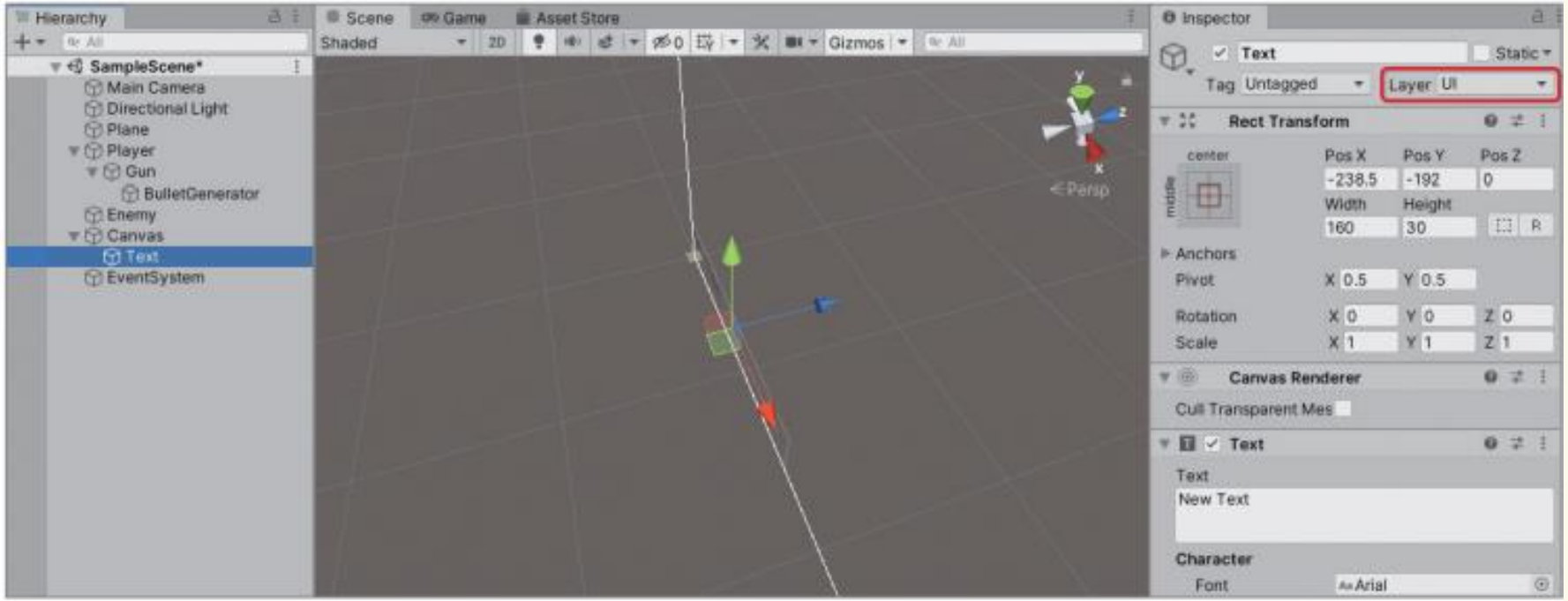
- 텍스트 UI 배치 예제
  - [Hierarchy] 창에서 +를 클릭하여 [UI]-[Text]를 선택
  - 자동으로 Canvas 오브젝트가 생기고 그 아래에 Text 오브젝트가 배치 -> Score





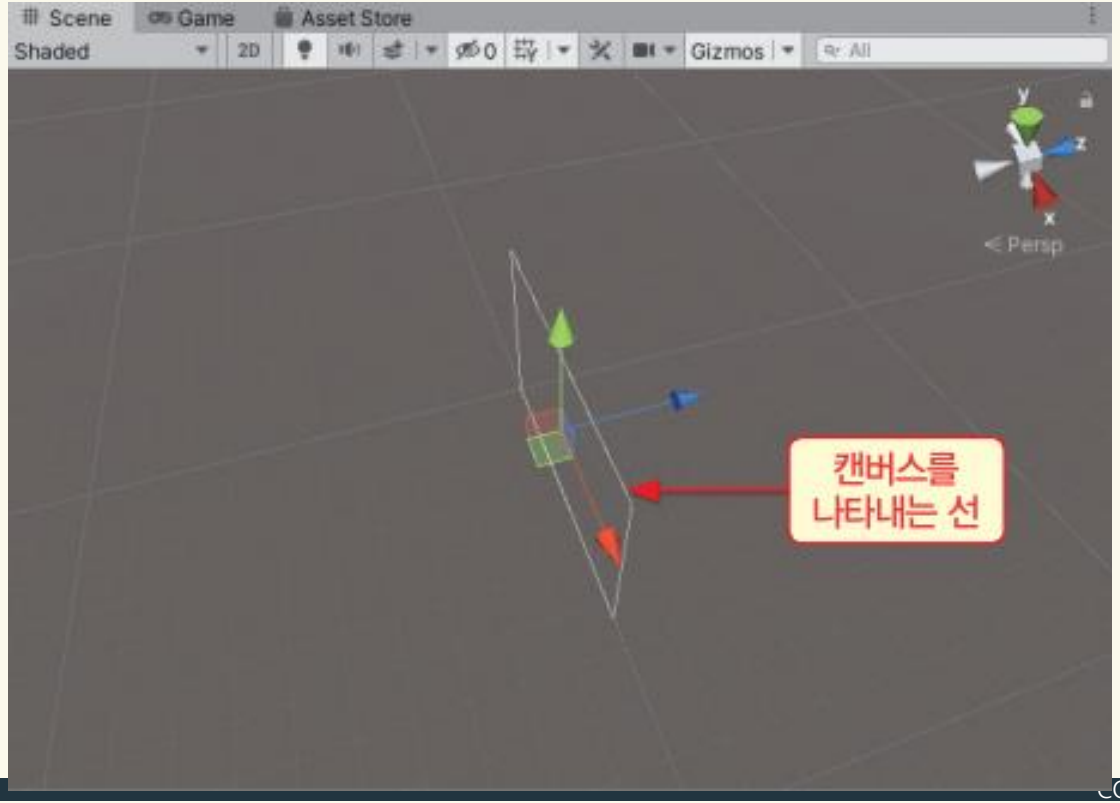
프리팹을 이용한 게임 제작

- Text 오브젝트를 더블클릭하면 [Scene] 창에 Text 오브젝트의 위치 표시



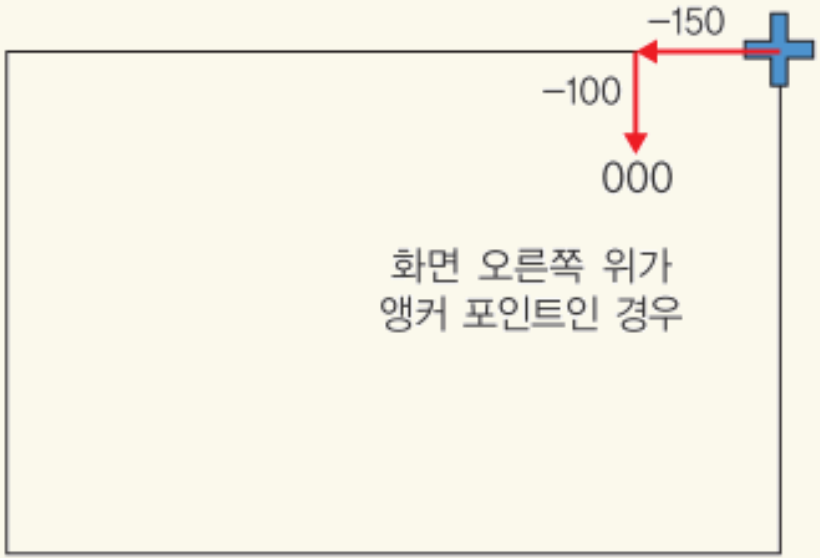
프리팹을 이용한 게임 제작

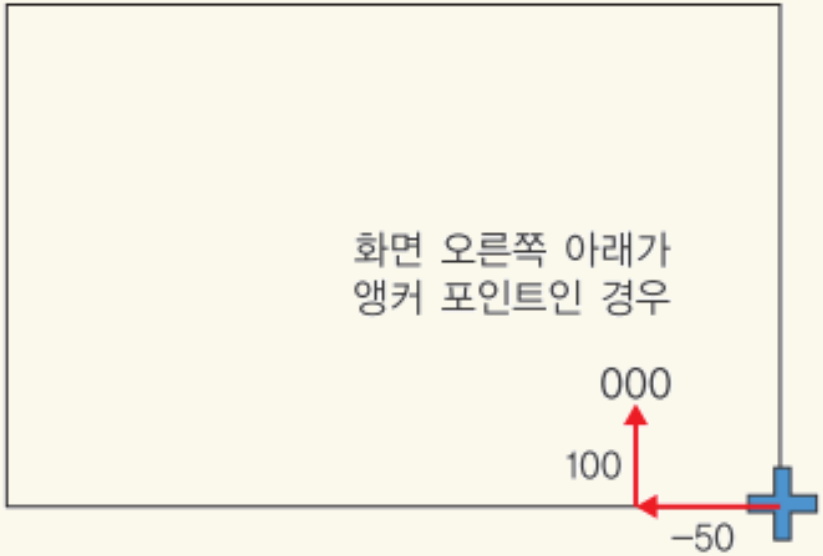
- Text 오브젝트가 위치한 영역 주위의 흰 선은 Canvas 오브젝트



프리팹을 이용한 게임 제작

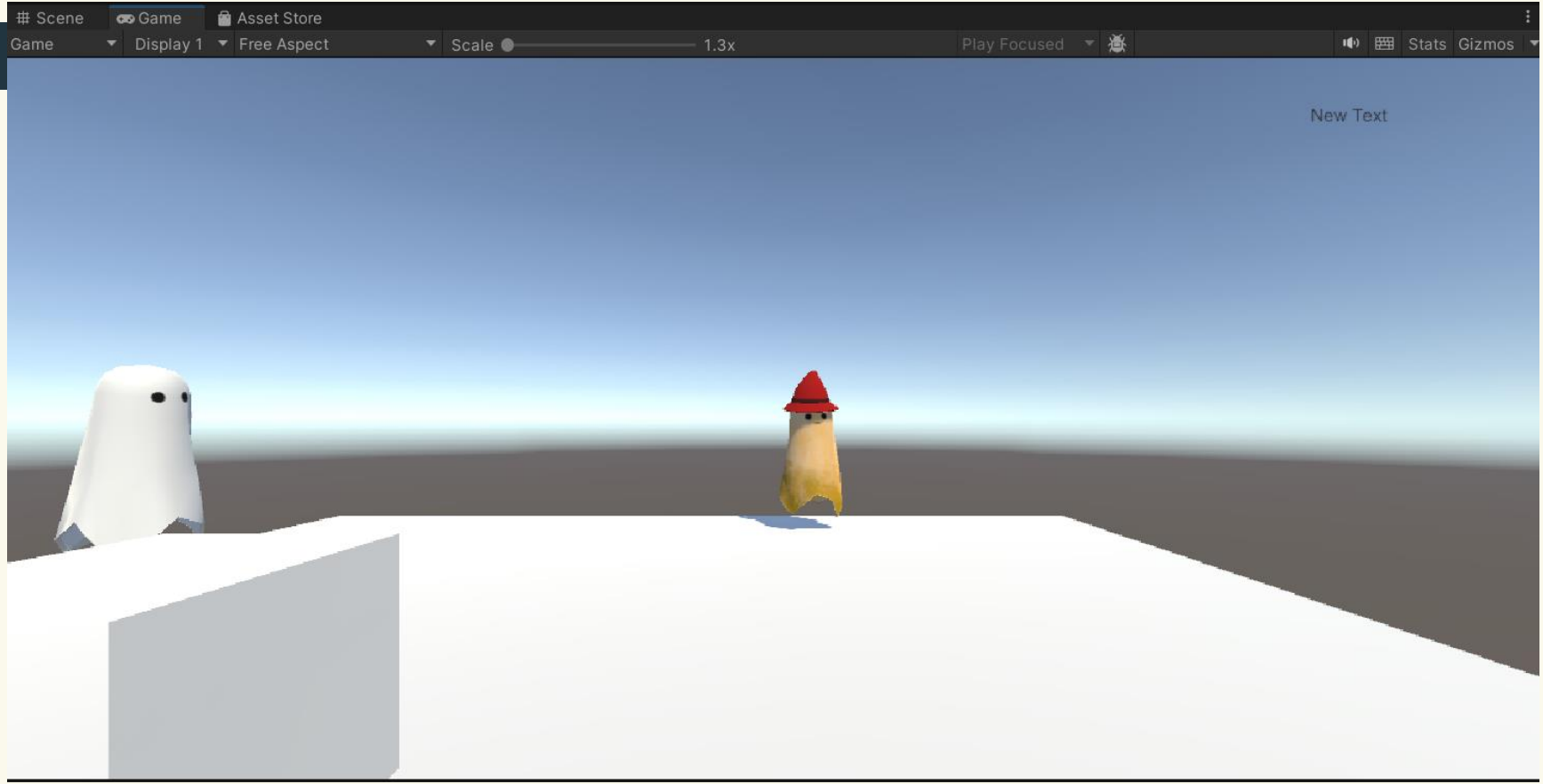
- 캔버스에서 텍스트가 표시될 위치를 조정하기
  - 앵커포인트: 위치 조정의 기준점



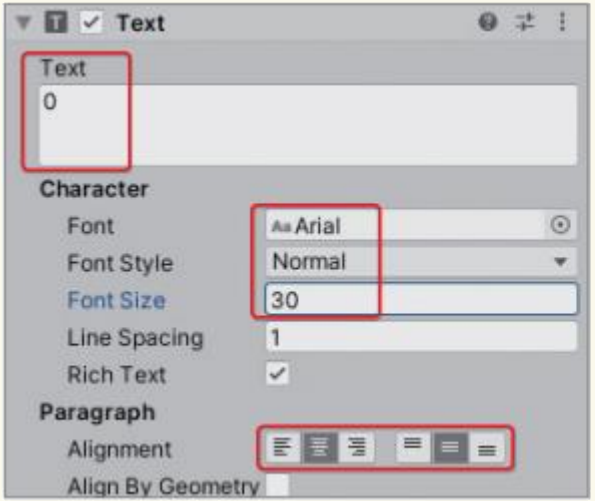


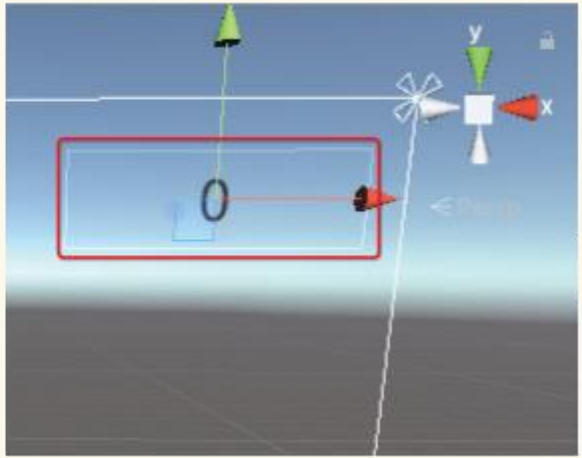
프리팹을

Inspector
Score
Static
Tag Untagged
Layer UI
Rect Transform
center
middle
Pos X
Pos Y
Pos Z
-238.5
-192
0
Width
Height
160
30
R
right
top
이 PC에 저장됨
Pos X
-692
Width
Anchor Presets
Shift: Also set pivot
Alt: Also set position
left
center
right
stretch
top
middle
bottom
stretch
Font Style
Normal
New Text
인스펙터창에서
앵커 포인트 선택
앵커포인트를
기준으로 x, y 위치와
크기 조정
Inspector
Score
Static
Tag Untagged
Layer UI
Rect Transform
right
top
Pos X
Pos Y
Pos Z
-100
-50
0
Width
Height
160
50
R
Anchors
Pivot
X 0.5
Y 0.5
Rotation
X 0
Y 0
Z 0
Scale
X 1
Y 1
Z 1
Canvas Renderer
Cull Transparent Mes
Text
Text
New Text
Character
Font
Arial


Scene
Game
Asset Store
Game
Display 1
Free Aspect
Scale
1.3x
Play Focused
Stats
Gizmos
New Text

프리팹을 이용한 게임 제작

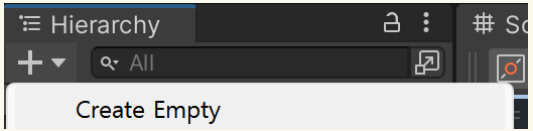


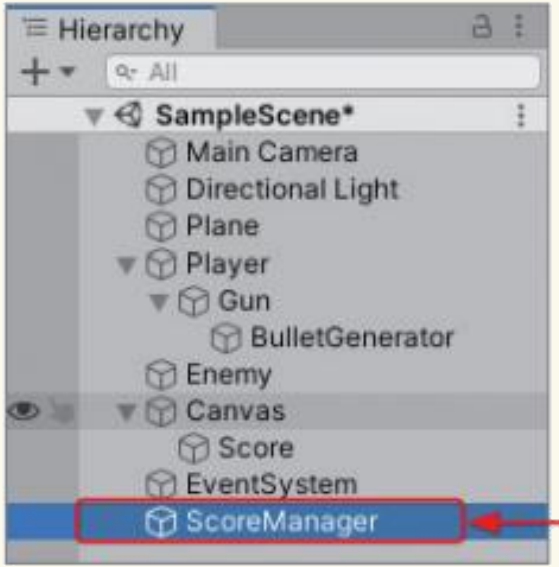


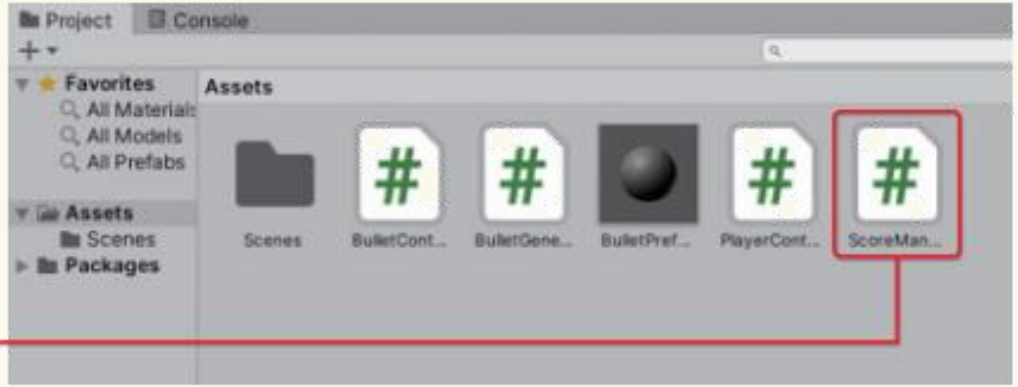
Text 항목에서 Score 오브젝트에 표시할 기본값, 폰트의 종류와 크기, 텍스트 정렬을 설정

프리팹을 이용한 게임 제작

- ScoreManager 스크립트를 적용할 빈 오브젝트 생성







프리팹을 이용한 게임 제작

- UI 스크립트 코딩하기: ScoreManager
  - UI와 관련된 코딩을 하려면 상단에 using UnityEngine.UI를 추가
  - UI 오브젝트를 찾을 때 GameObject.Find("찾을 오브젝트 이름") 함수를 사용

```
private Text score;
private int count;

void Start ()
{
    this.score = GameObject.Find("Score").GetComponent<Text>();
    count = 0;
}

public void IncScore ()
{
    count += 1;
    this.score.text = count.ToString ();
}
```

UI 오브젝트 찾기
점수는 0으로 초기화

숫자를 문자로 변환하여
text에 지정

- 스크립트 :

ScoreManager

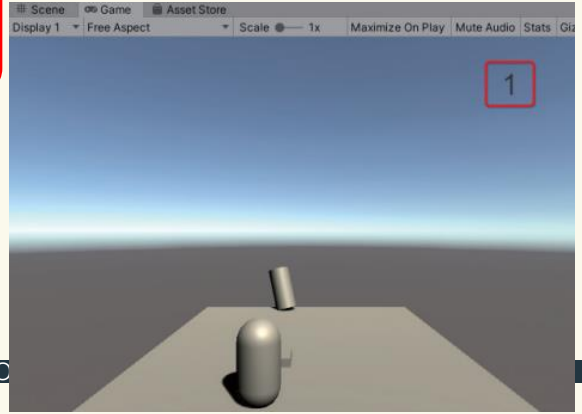
```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.UI;

public class ScoreManager : MonoBehaviour
{
    private Text score;
    private int count;

    // Start is called before the first frame update
    void Start()
    {
        this.score = GameObject.Find ("Score").GetComponent<Text> ();
        count = 0;
    }

    public void IncScore ()
    {
        count += 1;
        this.score.text = count.ToString ();
    }
}
```

프리팹을 이용한 게임 제작

- IncScore( ) 함수는 총알이 Enemy와 충돌할 때 호출해야 하므로 BulletController 스크립트의 OnCollisionEnter( ) 함수를 수정

```
void OnCollisionEnter (Collision coll)
{
    if (coll.collider.tag == "ENEMY")
    {
        GameObject manager = GameObject.Find ("ScoreManager");
        manager.GetComponent<ScoreManager> ().IncScore ();
        Destroy (gameObject, 0.2f);   // 오브젝트 파괴
    }
}
```

ScoreManager 오브젝트를 찾아서 IncScore() 함수 호출

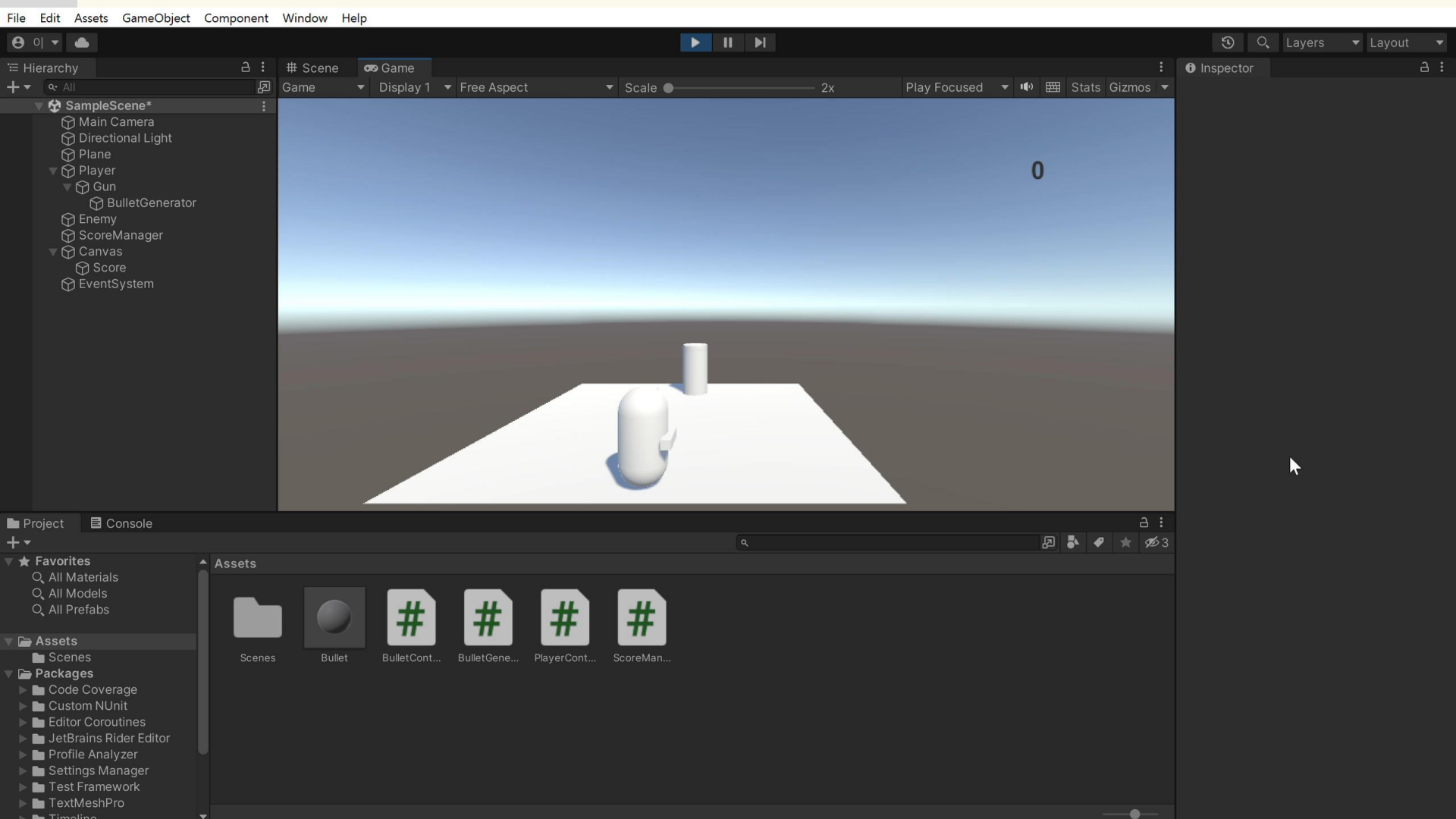


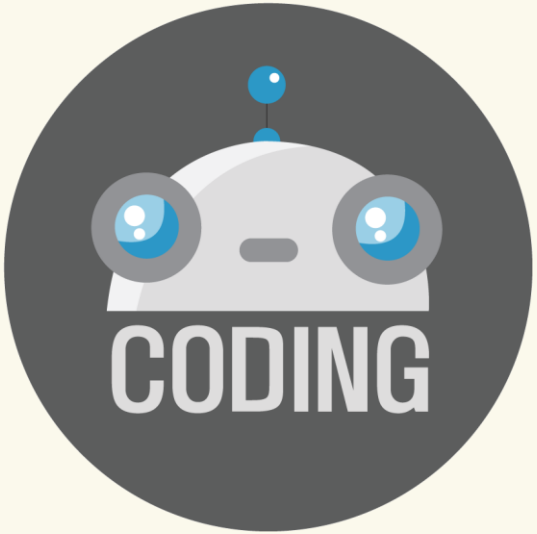
- 스크립트 :

BulletController

추가 수정

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
public class BulletController : MonoBehaviour
{
    void Start()
    {
    }
    void Update()
    {
    }
    // 총알 발사 함수
    public void Shoot (Vector3 dir)
    {
        GetComponent<Rigidbody> ().AddForce (dir);
    }
    void OnCollisionEnter (Collision coll)
    {
        if (coll.collider.tag == "ENEMY") {
            GameObject manager = GameObject.Find ("ScoreManager");
            manager.GetComponent<ScoreManager> ().IncScore ();
            Destroy (gameObject, 0.2f);   // 오브젝트 파괴
        }
    }
}
```


File
Edit
Assets
GameObject
Component
Window
Help
Layers
Layout
Hierarchy
SampleScene*
Main Camera
Directional Light
Plane
Player
Gun
BulletGenerator
Enemy
ScoreManager
Canvas
Score
EventSystem
Scene
Game
Game
Display 1
Free Aspect
Scale
2x
Play Focused
Stats
Gizmos
0
Inspector
Project
Console
Favorites
All Materials
All Models
All Prefabs
Assets
Scenes
Packages
Code Coverage
Custom NUnit
Editor Coroutines
JetBrains Rider Editor
Profile Analyzer
Settings Manager
Test Framework
TextMeshPro
Assets
Scenes
Bullet
BulletCont...
BulletGene...
PlayerCont...
ScoreMan...



THANK YOU